Local Search with Correlated Randomness

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Abstract

The running-time distribution of a randomized algorithm is computed under independent randomness, but independence is rarely what a source guarantees. We study sources ν satisfying $\nu[w] \le DP[w]^s$ for every finite prefix w of the random tape, where P is an independent reference law, $0 < s \le 1$, and $D \ge 1$; individual choices may be predictable, even determined by the past. The worst-case probability of any closed tape event E is exactly $\min\{1, D\mathcal{C}_s(E)\}$, where $\mathcal{C}_s(E)$ is the infimum cost of covering E by prefixes of varying depth, a tree max-flow/min-cut identity that makes worst-case survival probabilities exact and optimizable over the algorithm's choices.

Three applications follow. For n tasks on one machine, each retried until the next random symbol lies in its success set, every schedule has the same makespan distribution under independent randomness, yet under our constraint two fixed orders miss a linear deadline with worst-case probabilities 1 and $\exp[-\Theta(n)]$; a static priority rule is optimal at every deadline. For the standard local walk on a k-CNF with an assignment that satisfies a majority of each clause, we determine the sharp source threshold s_*. Below s_*, on some instances a single admissible source forces nontermination however clauses are selected; at s_*, with fixed D, the expected number of flips is at most cubic in the initial Hamming distance, against linear under independence, with matching worst-case instances; and any fixed margin above s_* restores linear dependence on the distance, with a constant inversely proportional to the margin. Third, verified backtracking for noisy predecessor search remains correct under a sufficiently strong bound on the comparison errors. In every setting the algorithm may also restart, which supplies no fresh randomness on a shared tape; we show that restarting restores exponential tails exactly when the nontermination event has a prefix cover of cost below one.

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1 Introduction

How much does an algorithm's running-time distribution under independent randomness reveal about its behavior when independence is no longer guaranteed? The distribution gives the probability that the algorithm is still running at each time, but not which early choices the surviving runs share. When later choices can depend on earlier ones, that missing structure matters, and it can matter decisively. This paper studies the question for an arbitrary algorithm whose only randomness comes from a single tape. We impose one cumulative constraint on the joint law of the algorithm's random choices: every initial segment of the tape has probability at most a constant times a fixed power of its probability under an independent reference law.

The source can be viewed as an adversary that sees the algorithm's state and every choice made so far, limited by one budget for the whole run rather than by a separate constraint at each step. We study this general source model in three local-search settings, scheduling, local SAT search, and noisy search, and for one decision available in all of them, when to restart. In each, the worst case over admissible sources can be computed exactly, and the answer changes what the algorithm should do: in scheduling it picks the order in which to attempt tasks, an order the running-time law under independent randomness cannot see; in local SAT search it gives a sharp source threshold, with worst-case time cubic rather than linear at the threshold; in noisy search it says how to verify answers under correlated errors; and in every setting it says exactly when restarting on the same tape restores exponential tails.

The source model. An algorithm consumes a tape of primitive random choices, one symbol at a time, from a finite alphabet. A prefix $w = a_1 \cdots a_t$ consists of the first t symbols of the tape, that is, of all random choices made through time t, and for a law ν on tapes, $\nu[w]$ is the probability that the tape begins with w. Let P be the reference law under which the symbols are independent with a common per-symbol law. We allow the tape to be generated by any law ν satisfying the prefix probability bound

$$\nu[w] \le DP[w]^s \quad \text{for every finite prefix } w, \qquad 0 < s \le 1, \quad D \ge 1.$$

The exponent s measures what fraction of the reference randomness is guaranteed, and the constant D allows a fixed amount of slack. For a uniform q-symbol reference, the condition says exactly that the first t symbols have min-entropy at least $st \log_2 q - \log_2 D$ for every t, where the min-entropy of a distribution is the negative base-two logarithm of its largest outcome probability, the standard measure of weak randomness [CG88]. Importantly, the assumption constrains the tape as a whole, bounding the entire sequence of choices up to each time and conditioning on nothing. Within a run, a source may therefore spend randomness early and then produce a run of symbols determined by what came before; across runs, the guarantee is never renewed, so a restart gives the algorithm no fresh randomness. The independent reference law prices these deviations and is the baseline against which the results are compared; s and D set the room for deviation, and the whole-prefix form decides how that room can be used (Section 2).

Why this class. Three properties make the class natural. First, it is general. A Chor–Goldreich source [CG88], a sequence of blocks each with min-entropy conditioned on all previous ones, renews its guarantee at every block, and the almost Chor–Goldreich sources of Doron, Moshkovitz, Oh, and Zuckerman [DMOZ23] relax that guarantee, in their basic form by letting a bounded number of blocks, at positions fixed in advance, be arbitrary. A Chor–Goldreich bound implies our condition, with D absorbing the block length, but our condition is a cumulative information constraint on

whole prefixes, not a per-block one: a run of determined symbols may begin at a history-dependent time and last as long as the randomness spent before it pays for. The adversarial source of Section 3, which forces zeros from the random moment the nth zero appears, is such a run, and so is the SAT index stream below, which repeats one index m times after $18m$ random ones. Second, it is exact: for every closed failure event, the worst-case probability over the class is attained and equals the cheapest cover of the event by prefixes of varying depth, a tree max-flow/min-cut identity (Section 2) that makes thresholds sharp and lets choices be optimized against the worst case itself. Third, it is global: the guarantee is not inherited by the subsequence that one component of an interleaved computation sees, nor renewed by a restart, so restarts, interleavings, and replays are analyzed, in all our theorems, under the single constraint on the original tape.

Scheduling retry tasks. The simplest setting is a scheduling problem: n *retry tasks* share one machine, each task is completed when the next random symbol falls in its success set and otherwise remains unfinished, and a *selector*, our word for any algorithm's decision rule, here the schedule, decides which unfinished task to attempt next, possibly using everything observed so far. Under independent symbols this is single-machine stochastic scheduling with independent geometric processing times and a makespan deadline [Pin22], and every selector has the same makespan distribution, because the makespan is the sum of the tasks' geometric numbers of attempts. It is also the simplest local search: each attempt is a local move that repairs one task and never undoes another, whereas a move of the SAT walk below can undo progress. Our question is what happens when the processing times are driven by one correlated tape. For an explicit reference law and source strength, two fixed orders miss a deadline linear in n with worst-case probabilities 1 and $\exp[-\Theta(n)]$. We then identify, for each source strength, a fixed order that is optimal at every deadline among all selectors, including history-dependent ones.

Local SAT search. The second setting is the standard local walk for SAT. Tompkins and Hoos [TH06] found such walks robust to several forms of derandomization but vulnerable to cyclic random streams; the following stream illustrates what our assumption permits and what our theorems control. Consider a 3-CNF formula for which some assignment makes at least two literals true in every clause; the walk is not told this assignment. Repeatedly select a violated clause and read an index from $\{1, 2, 3\}$ to choose which of its three variables to flip. A violated clause has all three literals false, so at least two of its variables disagree with the promised assignment, and at least two of the three choices move one step closer to it in Hamming distance. Under independent uniform indices, this drift finds a satisfying assignment in expected time linear in the initial distance [AGH17]. Initialization and clause selection can be deterministic.

Now fix a positive integer m and generate the indices as follows: $18m$ independent uniform indices, then the last of these repeated m times, then independent uniform indices forever. Once the first $18m$ indices have been observed, the next m indices are known. Every possible prefix of length t contains at least $18t/19$ independent draws, so its probability is at most $3^{-18t/19}$: the stream satisfies our assumption with $s = 18/19$ and $D = 1$. The number 18 is the smallest for which this exponent exceeds the source threshold of this walk, which Section 4 determines; our SAT theorem therefore guarantees termination, with an exponential tail, for this stream. Note that restarting the walk during the repeated block changes the assignment but does not restore fresh randomness: the next index is still known. Section 6 asks when restarting can help at all.

Noisy search. In the third setting the algorithm must not only finish but return the right answer. The problem is to find the predecessor of a target among sorted keys by comparisons, when each

comparison (a noisy label in active-learning terms) may be answered wrongly. The walk is on a binary search tree over the key intervals. At each node it asks, by majority over repeated queries, whether the target lies in the node's interval and on which side of the split; it descends to the chosen child when the check passes and backs up to the parent when it fails. The tape holds one error bit per comparison. Under the reference law the errors are independent with a fixed probability below $1/2$, and this verified backtracking [FRPU94] reaches the correct leaf after $O(\log n)$ queries with any fixed error probability. Under our condition a run of wrong answers can be certain given the past; we ask what the same walk guarantees under one correlated tape.

Approach. It helps to take the source's point of view. The source is a flow of probability mass down the tree of prefixes, and the constraint caps the mass through the node w at $DP[w]^s$. The source wins by steering mass into executions that fail, and the cheap way to do so is to bank randomness early and spend it later by forcing symbols when the algorithm is vulnerable. The tool is a *prefix cover*: a finite collection of prefixes such that every execution that has not succeeded by the deadline begins with one of them. The analysis may stop at a short prefix on one branch of the execution tree and follow another branch much further. The cost of a cover is $\sum_w P[w]^s$; since $\nu[w] \le DP[w]^s$, D times the cost bounds the worst-case failure probability.

The converse is the key fact: by the max-flow/min-cut theorem on the prefix tree, a cover is a cut, and the worst-case failure probability over all admissible sources is *exactly* $\min\{1, D\mathcal{C}_s(E)\}$, where $\mathcal{C}_s(E)$ is the infimum of the cover costs of the closed failure event E, an infimum that is attained when E is determined by finitely many symbols. The exactness is what makes optimization possible: minimizing cover cost over the algorithm's choices is the same as minimizing worst-case failure, and the minimum satisfies a Bellman recursion in which every value is capped at one, the normalized cost of covering the current prefix outright. All upper bounds in this paper are covers of this kind, and all lower bounds are sources built from the max-flow side of the identity.

What is technically new. The identity and the recursion are classical; the contributions are what they are made to yield: an exchange inequality compatible with the truncation at one, which reduces a dynamic program over 2^n task subsets to the n suffixes of a sorted list (Theorem 3.1); a barrier lemma for integer potentials under a worst-case count of favorable symbols (Lemma C.1); a discrete capped plateau with near-critical slack, from which one source forcing the matching delay for every selector is built (Lemma C.3, Theorem 4.3); and the impossibility half of the restart criterion, against all history-dependent policies (Theorem 6.1).

An optimal retry order that running-time laws cannot identify. For a uniform reference source, the optimal rule is to finish a task with the fewest successful symbols first, that is, largest expected length first; for a general reference law, Section 3 replaces this count by a weight of the success set. The source stalls a task by forcing symbols outside its success set, and forcing a common symbol is cheap under the prefix bound while forcing a rare one is expensive; so the task that needs a rare symbol is the one the source can stall most cheaply, and it should be attempted before the source has banked slack from earlier, more random, symbols. Always attempting an unfinished task of minimum weight minimizes worst-case failure at every deadline and every D, over all deterministic history-dependent selectors (Theorem 3.1). Neither the running-time law under P nor the sum of s-th powers of the surviving-prefix probabilities at any single depth can distinguish these selectors: both statistics are the same for all of them. The opening exponential separation between the best and worst orders is Theorem 3.2; its proof is a cover that cuts some executions at time $4n$ and others at time $8n$.

A sharp SAT threshold and matching delay. Take the walk above on a k-CNF for which some assignment makes at least r literals true in every clause, $k/2 < r < k$. Under independent uniform choices it takes $O_{k,r}(h)$ expected flips from Hamming distance h. Under the prefix bound, the sharp source threshold over this promise class is $s_* = \log_k(2\sqrt{r(k-r)})$. At and above it,

$$\mathbb{E}_\nu \tau = O_{k,r}\Big(\min\{L^3, L/(s-s_*)\}\Big), \qquad L = h + \log D + 1, \quad L/0 = \infty,$$

and at the threshold the probability of surviving T flips is at most $\exp[-\Omega_{k,r}(T^{1/3})]$ for fixed h and D, a stretched-exponential bound that a complete formula family matches for deadlines up to the cube of the number of variables. The upper bound allows arbitrary clause overlap and history-dependent clause selection. On suitable formulas and initial assignments, one source forces the matching delay with probability one, simultaneously for every selector. Thus, at the threshold and fixed D, a walk whose independent-source expectation is linear in the initial distance h can be forced to take time cubic in h (Theorems 4.1 and 4.3); on these instances $n \asymp h^3$, so in terms of n the slowdown is from $n^{1/3}$ to n. A fixed margin above the threshold restores linear dependence on L. Below s_*, nontermination is possible even with $D = 1$ (Theorem 4.2).

Correct output as well as termination in noisy search. Under the prefix bound on comparison errors, at any source strength at or above a threshold s_c that we show suffices but do not show is sharp, we give a predecessor-search algorithm with error probability at most η and $O_p((h + \log(D/\eta) + 1)^3)$ expected comparisons, where h is the correct leaf's depth and the algorithm needs neither h nor the tree height. Longer verification chains at deeper leaves place all wrong outputs beyond one common distance barrier, so a single cover controls the total error without a union bound over leaves. A fixed margin above this sufficient threshold gives linear dependence on the same quantity (Theorem 5.1).

An exact criterion for restarting on the same tape. A restart changes the state but supplies no fresh randomness. Nevertheless, for a fixed algorithm, restarting can give exponential tails exactly when its nontermination event has a prefix cover of cost $c < 1$, and the same condition characterizes whether any restart rule can terminate almost surely against every source with $D = 1$ (Theorem 6.1): the cover tells the algorithm which attempts to abandon, and j abandoned attempts in a row form a cover of cost c^j. For explicit finite-state processes, two linear programs, one for the values and one that breaks ties among value-optimal updates, compute a stationary selector minimizing worst-case nontermination and a restart rule whenever robust termination is possible (Theorem 6.2). The rule must be adaptive: a three-state process has worst-case nontermination probability at least $1/2$ under every predetermined cutoff sequence, while one stationary rule gives an exact exponential tail (Proposition D.1). For SAT, an observable rule resets on leaving a Hamming ball around the initial assignment; it needs a distance bound but not the satisfying assignment, and replay removes the supplied bound (Theorem 4.4 and Appendix C.6).

Related work. Computation with weak randomness [CG88, DMOZ23, DMOZ25] asks how to extract near-uniform bits from a defective source, or to condense it into a shorter string with more randomness per symbol, and then to run a randomized algorithm on the result; we instead take the algorithm as given, let it read the defective source directly, and determine exactly how it fares. Extracting first is not the question here: the model supplies no independent seed, and the prefix condition alone admits no generic deterministic conversion to near-uniform bits, since the class contains the semi-random sources of Santha and Vazirani [SV86], in which each bit may be biased given the past and from which no deterministic procedure produces a less biased bit. The classical ingredients, the tree max-flow/min-cut theorem behind Frostman's lemma, affine monotone dynamic

programming, and the cube-root scale of branching processes with an absorbing wall, are credited where they are used. Variable-length prefixes appear as a proof device in the entropy-compression analysis of Achlioptas, Iliopoulos, and Vlassis [AIV17]. The settings are inherited, and what is new is their analysis under one correlated tape: the scheduling setting is classical single-machine stochastic scheduling [Pin22]; the SAT walk is that of Austrin, Guruswami, and Håstad [AGH17]; verified backtracking is that of Feige, Raghavan, Peleg, and Upfal [FRPU94]; and restart schedules for independent runs go back to Luby, Sinclair, and Zuckerman [LSZ93].

Organization. Section 2 gives the model and the capacity identity; Sections 3–5 treat the three settings and Section 6 the restart decision; Section 7 asks further questions. Related work is detailed in Appendix A, and all proofs are in the appendices.

2 Prefix covers and source robustness

Tapes and sources. Fix a finite alphabet $\mathcal{A}$ with at least two symbols and a full-support distribution π on it, and let $P = \pi^{\mathbb{N}}$ be the product law. A *tape* is a sequence in $\mathcal{A}^{\mathbb{N}}$, and a *word* is a finite sequence $w \in \mathcal{A}^*$; w is a *prefix* of a tape ω if ω begins with w. The *cylinder* $[w]$ is the set of tapes with prefix w, and for a law ν on tapes, $\nu[w] = \nu([w])$ is the probability that w is a prefix of the tape; under the reference law, $P[w] = \prod_i \pi(w_i)$. The symbols are the actual primitive choices consumed by the algorithm, and their alphabet is fixed once and for all: grouping or recoding symbols changes the source assumption.

For $0 < s \le 1$ and $D \ge 1$, define the source class

$$\mathcal{Q}_{s,D} = \{\nu : \nu[w] \le DP[w]^s \text{ for every word } w\}. \tag{1}$$

Here ν ranges over probability laws on whole tapes, and one constant D applies to prefixes of all lengths; we refer to (1) as the *prefix probability bound*, or the prefix bound for short. No randomness is reserved for the algorithm outside this tape. Larger s strengthens the guarantee, whereas larger D weakens it. At $s = D = 1$ the class contains only P. Equivalently, (1) is a cumulative information constraint: along every path, the increments $\log \nu[wa] - \log \nu[w] - s \log \pi(a)$ of the log-likelihood ratio against the reference raised to the power s sum to at most $\log D$. The increments may be negative, so randomness left in the tape early is banked and can be drawn on later.

The class is strictly larger than the class of sources with a conditional guarantee. For a uniform q-symbol reference, a bound of q^{-s} on the conditional probability of every next symbol implies membership in $\mathcal{Q}_{s,1}$, but the ternary stream of the introduction belongs to $\mathcal{Q}_{18/19,1}$ and has conditional probability one for m consecutive symbols. In the other direction, conditioning on a prefix w of positive ν-probability gives $\nu[v \mid w] \le D_w P[v]^s$ with $D_w = DP[w]^s/\nu[w]$, which can be much larger than D. There is no uniform reset of the guarantee after a history; this is why restarts supply no fresh randomness. Appendix B records further examples and explains why generators with a finite seed lie outside the model, as they do outside every class with a positive entropy rate.

Selectors, termination, and covers. A *selector* is the algorithm's decision rule, a deterministic policy in the language of control. It is *nonanticipating*: at each step it may use the prefix of the tape read so far, which determines the entire history of states and actions, but no future symbols. A prefix is *live* if the algorithm has not terminated after reading it. Let τ be the number of updates the algorithm makes, including the update on which it terminates, so that $\{\tau > T\}$ is the event that the algorithm is still live after T updates. Write $\mathcal{N} = \{\tau = \infty\}$ for the nontermination event.

For a closed tape event E (closed in the product topology; every event determined by finitely many symbols is closed, and so is $\mathcal{N}$), define its *prefix capacity*

$$\mathcal{C}_s(E) = \inf_{\mathcal{B}} \sum_{w\in\mathcal{B}} P[w]^s, \qquad E \subseteq \bigcup_{w\in\mathcal{B}} [w], \tag{2}$$

where $\mathcal{B}$ ranges over finite prefix-free covers. Prefix-free means that no chosen word extends another, so the cylinders are disjoint; their lengths may differ. The empty word alone is a cover of cost one. Every cover gives an upper bound on $\nu(E)$ for every $\nu \in \mathcal{Q}_{s,D}$, by summing the source bounds over its words. The identity below gives the converse: some admissible source attains the best such bound.

Theorem 2.1 (Classical tree capacity identity)**.** *For every closed E, $0 < s \le 1$, and $1 \le D < \infty$,*

$$\max_{\nu\in\mathcal{Q}_{s,D}} \nu(E) = \min\{1, D\mathcal{C}_s(E)\}. \tag{3}$$

If E depends on the first T symbols, a minimizing cover uses words of length at most T. For events E_T that depend on finitely many symbols and decrease to $\mathcal{N}$,

$$\max_{\nu\in\mathcal{Q}_{s,D}} \nu(\mathcal{N}) = \lim_{T\to\infty} \max_{\nu\in\mathcal{Q}_{s,D}} \nu(E_T). \tag{4}$$

A source is a flow of probability mass down the prefix tree, in which the node w has capacity $DP[w]^s$, and a cover is a cut separating the root from every branch in E. The identity is the tree max-flow/min-cut theorem, a minimax duality between sources and covers, as in the tree proof of Frostman's lemma [BP17, Section 3.1]. The one additional step, completing a maximal flow into E to a probability law on all tapes that still respects every capacity, is carried out in Appendix B. In the metric $d_P(\omega,\omega') = P[\omega \wedge \omega']$ on tapes, where $\omega \wedge \omega'$ is the longest common prefix, $\mathcal{C}_s(E)$ is the s-dimensional Hausdorff content of E, and the Hausdorff dimension of the nontermination set is the threshold in s below which some finite-D source is supported on it (Corollary B.1); no later result uses this.

The cover recursion. Fix a deadline T, one symbol per update, and the event $E = \{\tau > T\}$. For a prefix w, let $V(w) = \mathcal{C}_s(E\cap[w])/P[w]^s$ be the cost of covering the part of E below w, measured in units of the cost $P[w]^s$ of covering $[w]$ outright; thus $V(w) \le 1$, $V(w) = 1$ for a live word of length T, and $V(w) = 0$ for a terminated word. Working backwards, at a prefix w one either covers everything below it, at normalized cost one, or covers its children separately, so

$$V(w) = \min\left\{1, \sum_{a\in\mathcal{A}} \pi(a)^s V(wa)\right\}, \tag{5}$$

and the root value is $\mathcal{C}_s(E)$. Different branches can make different choices, whereas a single-depth sum forces every branch to stop at the same time. When the algorithm also chooses an update at each step, the recursion takes a minimum over updates as well, and it becomes a Bellman equation whose optimal policy is the robust algorithm (Proposition B.2).

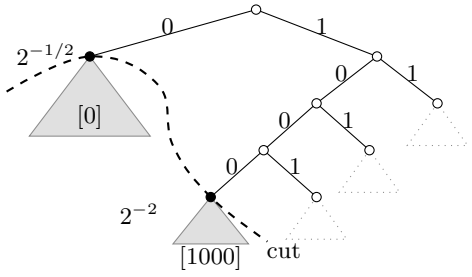


Figure 1: Fair bits, $s = 1/2$, $E = [0]\cup[1000]$ (shaded). The cover $\{0, 1000\}$ (filled nodes) is a cut separating the root from E, of cost $2^{-1/2}+2^{-2} < 1$; every single-depth cover costs at least one.

Figure 1 gives a minimal example: with fair bits and $s = 1/2$, the cover $\{0, 1000\}$ of $E = [0]\cup[1000]$ costs $2^{-1/2} + 2^{-2} < 1$, whereas depths $0, \dots, 4$ cost $1, \sqrt{2}, 3/2, 5/\sqrt{8}, 9/4$ and refining further only increases the cost: even the best single depth loses information.

What a single-depth bound retains. A single-depth cover uses the words of one length or of one update count. With $\mathcal{L}_T$ the minimal prefixes completing T updates, including a terminating one, its cost is $Z_T(s) = \sum_{w\in\mathcal{L}_T} P[w]^s$. Then $Z_T(1) = P\{\tau \geq T\}$ and $\mathbb{P}_\nu\{\tau \geq T\} \leq DZ_T(s)$; with one symbol per update, $\widetilde{Z}_T(s)$ is the same sum over the length-T words still live, which bounds $\{\tau > T\}$. We call $Z_T(\theta)$ and $\widetilde{Z}_T(\theta)$, for any exponent $\theta > 0$, the *prefix power sums* at depth T. For a uniform q-symbol source, $Z_T(s) = q^{(1-s)T}P\{\tau \geq T\}$, the independent-source tail inflated by $q^{(1-s)T}$. A prefix cover can combine levels, and therefore uses how the surviving prefixes extend one another.

3 Scheduling retry tasks: an optimal order the stopping-time law cannot see

The model is deliberately simple: attempting one task does not change the others, so under independent randomness the order is irrelevant, whereas under the prefix bound it determines whether a linear deadline is missed with probability one or with exponentially small probability.

The model. There are n unfinished tasks and a finite symbol alphabet $\mathcal{A}$. Task i has a nonempty success set $B_i \subseteq \mathcal{A}$. An attempt chooses an unfinished task and reads the next symbol. The task completes if the symbol lies in B_i; otherwise it remains unfinished. Completed tasks are never selected again. The success sets may overlap. A deterministic nonanticipating selector may use the entire observed history. Let τ count attempts until all tasks complete. For $P = \pi^{\mathbb{N}}$ with full-support π, put

$$W = \sum_{a\in\mathcal{A}} \pi(a)^s, \qquad w_i = \sum_{a\in B_i} \pi(a)^s.$$

Theorem 3.1 (Optimal priority under a prefix probability bound)**.** *For every $0 < s \leq 1$, always attempting an unfinished task with smallest w_i minimizes $\max_{\nu\in\mathcal{Q}_{s,D}} \mathbb{P}_\nu\{\tau > T\}$ over all deterministic nonanticipating selectors, simultaneously for every $T \geq 0$ and $D \geq 1$. If $w_1 \leq \cdots \leq w_n$, set $v_t(n+1) = 0$, $v_0(j) = 1$ for $1 \leq j \leq n$, and*

$$v_{t+1}(j) = \min\{1, (W - w_j)v_t(j) + w_j v_t(j+1)\}.$$

The exact optimum is $\min\{1, Dv_T(1)\}$. Once the weights are computed and sorted, this takes $O(nT)$ arithmetic operations and $O(n)$ memory. Every selector has the same stopping-time law under P and the same prefix power sums $Z_T(\theta)$ and $\widetilde{Z}_T(\theta)$ at every depth T and every exponent $\theta > 0$.

For a uniform reference source, the rule is to finish a task with the fewest successful symbols first. It is an index policy, each task having the fixed index w_i and the smallest index being served first, in the tradition of priority policies for stochastic scheduling [MSU99, Pin22]. For a nonuniform reference, the weight w_i of the success set replaces this count. It is a cover weight rather than a probability: all symbol weights sum to W, which may exceed one. With singleton success sets, the rule still orders tasks by increasing ordinary success probability; in general the order can depend on s.

The recursion in the theorem is the cover recursion (5) on the state j, the index of the first unfinished task in sorted order: an attempt at task j fails at cost $W - w_j$ times the value with the same tasks remaining, or succeeds at cost w_j times the value with one fewer task. The content of the theorem is that nothing is gained by attempting tasks in any other order, or by switching orders in response to the observed history. Appendix E.1 proves this by a weighted exchange inequality that is compatible with the truncation at one; the exchange inequality is the step that reduces a dynamic program over all 2^n subsets of unfinished tasks to the n suffixes of the sorted list.

A quantitative separation. Take n tasks of type A, successful on 0, and one task of type B, successful on 1. Write Λ_A for the order that completes all A tasks first and Λ_B for the order that completes B first.

Theorem 3.2 (Equal prefix sums, different robustness)**.** *Fix $P(0) = 15/16$, $P(1) = 1/16$, and $\mathcal{Q} = \mathcal{Q}_{1/2,1}$. For every $n \geq 1$,*

$$\max_{\nu \in \mathcal{Q}} \mathbb{P}_\nu^{\Lambda_A}\{\tau > 8n\} = 1, \tag{6}$$

$$\left(\frac{15}{16}\right)^{4n} \leq \max_{\nu \in \mathcal{Q}} \mathbb{P}_\nu^{\Lambda_B}\{\tau > 8n\} \leq \frac{9}{8}\left(\frac{225}{256}\right)^{n} < 1. \tag{7}$$

One source $\nu_n \in \mathcal{Q}$ gives survival probability one for Λ_A and at most the upper bound in (7) for Λ_B. Moreover, the lower bound in (7) holds for every selector, Λ_B is minimax optimal at every deadline, and all deterministic nonanticipating selectors have identical stopping-time laws under P and identical prefix power sums at every depth and every exponent $\theta > 0$.

Since Λ_B attempts the task of smallest weight first, Theorem 3.1 gives its optimality and the equal laws and power sums; the lower bound holds for every selector because its proof uses a source whose first $8n$ symbols are all zeros with probability $(15/16)^{4n}$.

Why all power sums agree. The prefix power sums with exponent θ are, up to the factor $(\sum_a \pi(a)^\theta)^T$, survival probabilities under the tilted independent law with symbol probabilities proportional to $\pi(a)^\theta$. Under any independent law, preassign an independent stream of symbols to each task: the next unread symbol of any stream has the same law whatever the selector has observed, so the total number of attempts is the sum of the tasks' independent geometric workloads, whichever stream the selector reads next (Appendix E.1).

Why the robust tails differ. The bad order Λ_A postpones the only task that needs a 1. An adversarial source exploits this by generating zeros with probability $3/4$ until the nth zero appears, then forcing zeros through time $8n$, and then following P. To check the source constraint, track the ratio $\nu[w]/P[w]^{1/2}$ along the tape, which must stay at most one. A random one leaves it unchanged, each random zero multiplies it by $(3/4)/(15/16)^{1/2} < 1$, and each forced zero multiplies it by $(16/15)^{1/2} > 1$; the probability $3/4$ is chosen so that the n random zeros outweigh the at most $7n$ forced ones. Under Λ_A, either some A task is still unfinished at time $8n$, or the A tasks finish first and task B then sees only zeros for the rest of the deadline.

For the good order Λ_B, no single depth gives a useful bound: Λ_A survives with probability one, so every single-depth cover of its survival event costs at least one, and by Theorem 3.1 the same sums are those of Λ_B. A cover at two depths does better. Cover first by the prefix 0^{4n}, on which B is still unfinished at time $4n$. Otherwise the first 1 completes B by time $4n$, and survival to time $8n$ requires fewer than n zeros among the remaining symbols; a generating-function bound covers these paths at their full depth. The resulting mixed-depth cover proves (7), and Appendix E.2 checks the source constraint and both bounds.

Thus the cover resolves an algorithmic choice that every single-depth power sum leaves open. One tape shared by several independent processes is likewise delicate: Appendix G gives processes, each robust alone, that one admissible shared source keeps from jointly terminating under every interleaving.

4 Local SAT search: sharp bounds and restarts

A flip can satisfy one clause while violating several others, so the number of violated clauses is not a reliable measure of progress for a local walk. The promise below supplies a different measure: the Hamming distance to an unknown assignment that satisfies a majority of each clause's literals. The walk never sees this distance, but every flip it makes changes it by exactly one.

The promise and the walk. Fix integers $k/2 < r < k$. A k-CNF is *strongly r-satisfiable* if some assignment x^* makes at least r literals true in each clause; clauses use k distinct variables. We call such an x^* a strong witness. The walk need only find an ordinary satisfying assignment, not a strong witness. At each step the algorithm chooses any violated clause and flips the variable indexed, under a fixed ordering of its literals, by the next source symbol, which is uniform on $[k]$ under the reference law P; the clause choice may depend on the complete past. This is an established polynomial-time algorithm for the promise problem [AGH17, Proposition 6.1]: when a clause is violated, all of its literals are false, so every literal that x^* satisfies sits on a variable where the current assignment and x^* disagree, and at least r of the k possible flips decrease the distance to x^*. Put

$$\lambda = \sqrt{\frac{r}{k-r}} > 1, \qquad s_* = \log_k(2\sqrt{r(k-r)}) \in (0,1). \tag{8}$$

Theorem 4.1 (Robustness from distance to a solution)**.** *Let Φ be a strongly r-satisfiable k-CNF on n variables, and let h be the Hamming distance of its initial assignment from a strong witness x^*. Put $L = h + \log D + 1$ and $\delta = s - s_*$. There are constants $c, C > 0$, depending only on k, r, such that for every deterministic nonanticipating clause selector, $s \in [s_*, 1]$, $D \geq 1$, $\nu \in \mathcal{Q}_{s,D}$, and integer $T \geq 0$,*

$$\mathbb{P}_\nu\{\tau > T\} \leq \min\{1, CD\lambda^h e^{-cT^{1/3}}, D\lambda^h k^{-\delta T}\}, \tag{9}$$

$$\mathbb{E}_\nu \tau \leq C \min\{L^3, L/\delta\}. \tag{10}$$

We interpret $L/0 = \infty$. Here τ counts variable flips until the formula is satisfied; there is no restriction on clause overlap.

Under independent randomness the walk's expected flip count is at most $kh/(2r-k)$, by the drift of the distance. The robust bound depends on the distance to a strong witness and not on the number of variables, because the proof pays only for excursions in the distance. The statement counts flips; finding a violated clause takes additional work.

Why the threshold is s_*. At a violated clause, at least r of the k labels (next symbols) decrease the distance d to x^* by one and the rest increase it by one; the distance process need not be Markov, and in a cover sum each label carries the weight k^{-s}. To bound the cover recursion by a function $\phi(d)$ of the distance alone, nondecreasing in d, it suffices that $\phi(d) \geq a_s\phi(d-1) + b_s\phi(d+1)$ wherever the recursion continues: extra decreasing labels only lower the children's sum, so the inequality propagates backward, where

$$a_s = rk^{-s}, \qquad b_s = (k-r)k^{-s}.$$

These are weights in a cover sum, not conditional probabilities of the actual next move. Dividing $\phi(d)$ by λ^d turns this two-sided comparison into a symmetric one with total weight $2\sqrt{a_s b_s}$ per step; the threshold s_* is where this total weight equals one. At that point $a_{s_*} = \lambda/2$ and $b_{s_*} = 1/(2\lambda)$, so after the change of scale the comparison is a symmetric simple random walk, with its diffusive time scale. Above the threshold the total weight is $k^{-\delta} < 1$ per step, which gives the exponential tail in (9).

Why different depths help. At the threshold, a single-depth cover decays only polynomially (Theorem 4.2). The stretched-exponential tail comes from covering at two kinds of depth. Choose a barrier $m > h$. Cover a surviving run when it first reaches distance m, or at time T if it has stayed below m throughout. The first part costs at most $(h/m)\lambda^{h-m}$, because the function $(d/m)\lambda^{d-m}$ is harmonic for the two-sided comparison and equals one at the barrier; so the source can give a distant excursion only exponentially small probability in m. The second part costs at most $C\lambda^h e^{-cT/m^2}$, because a symmetric walk confined to a strip of width m decays on time scale m^2. Balancing m against T/m^2 gives $m \asymp T^{1/3}$ and the stated tail. The first-hit prefixes have different lengths even though there is only one spatial barrier.

A reusable barrier principle. Nothing in this argument is specific to SAT: the distance is a Lyapunov potential whose drift condition is replaced by a worst-case count of favorable symbols. Lemma C.1 in Appendix C states it for any deterministic process driven by symbols in $[k]$: assign a positive integer potential to every live history, with potential zero on termination, and assume that at every live history at least r of the k next symbols either terminate or decrease the potential by at least one, while every other symbol either terminates or increases it by at most one. Then, for every $\nu \in \mathcal{Q}_{s_*,D}$ and every barrier m above the initial potential h, the two-part bound above holds with $\lambda = \sqrt{r/(k-r)}$: survival at time T has probability at most $\min\{1, D[(h/m)\lambda^{h-m} + C\lambda^h e^{-cT/m^2}]\}$, and the expected running time is $O(L^3)$ with $L = h + \log D + 1$. The potential need not be Markov, the state space need not be finite, and blocks of symbols with nonuniform weights are allowed, as in Section 5. For comparison, a single-depth argument that reflects at the global distance bound n gives only $\mathbb{E}_\nu \tau = O_{k,r}(n^2(h + \log D + 1))$ at $s = s_*$; the barrier buys h^2 in place of n^2, and the critical tail. The barrier sits at an unknown distance; an observable Hamming ball replaces it below.

The threshold is sharp. The complete formula family Φ_n of the theorem below, every clause with r positive and $k - r$ negative literals on distinct variables, supplies exact lower bounds. A clause of Φ_n is violated exactly when its r positive variables are zero and its $k - r$ negative variables are one, so an assignment with d zeros is live exactly when $r \le d \le n - k + r$. The walk is therefore a one-dimensional process on these $N - 1 = n - k + 1$ live distances, absorbed at both ends, and its transitions are the same for every clause selector.

Theorem 4.2 (Sharp threshold and a gap from every single depth)**.** *Let Φ_n contain all clauses with r positive and $k - r$ negative literals on distinct variables, with positive literals ordered first, and put $N = n - k + 2 \ge 3$.*

(i) For every $0 < s < s_*$, some n, initial assignment, and single source in $\mathcal{Q}_{s,1}$ force nontermination under every nonanticipating clause selector.

(ii) Start with exactly r zero variables. Uniformly in $s \in [s_*, 1]$, $N \ge 3$, and $T \ge 1$, the worst survival probability is

$$A_{N,T}(s) := \max_{\nu \in \mathcal{Q}_{s,1}} \mathbb{P}_\nu\{\tau > T\} = \exp\Big[-\Theta_{k,r}\Big(T^{1/3} + (s - s_*)T + T/N^2\Big)\Big]. \tag{11}$$

At $s = s_*$, for the same event $E_T = \{\tau > T\}$, let $F_t(E_T)$ be the minimum cost of a cover using only length-t cylinders. Then

$$\inf_{t \ge 0} F_t(E_T) = \Theta_{k,r}((T+1)^{-3/2}) \qquad (1 \le T \le N - 2). \tag{12}$$

Thus s_* is the sharp uniform threshold for this walk over the promise class; the three terms in (11) are the critical cube-root tail, the gain from a margin above the threshold, and the gain from the global distance bound n. At $s = s_*$ and the deadline $T = n - k$, variable-depth covers give $e^{-\Theta(n^{1/3})}$, while every single-depth cover of the same event costs at least order $n^{-3/2}$. The symmetric family supplies the lower bounds only; Theorem 4.1 requires no symmetry.

For $k = 3, r = 2$, $s_* = \log_3(2\sqrt{2}) \approx 0.9464$, and 18 is the least number of fresh draws per repeated draw in the introductory stream whose exponent, $18/19 \approx 0.9474$, exceeds it. For $k = 3, r = 2$ a strong witness can be found by 2-SAT, replacing each clause by its three pairwise subclauses, so the result concerns the robustness of the specified walk, not tractability. The obstruction below s_* need not cycle: Corollary C.2 gives a full-support source with failure probability at least $1 - \varepsilon$ for every selector, under which the trajectory of any memoryless selector is almost surely not eventually periodic.

The whole transition is sharp. Define

$$\Psi_s(L) = \min\{L^3, L/(s - s_*)\}, \qquad L/0 = \infty.$$

Theorem 4.3 (Matching delay across the source threshold)**.** *Fix $k/2 < r < k$. There is a constant $c_{k,r} > 0$ such that for every $s \in [s_*, 1]$, integer $h \geq r$, and $D \geq 1$, some complete formula Φ_n from Theorem 4.2 has an initial assignment that does not satisfy Φ_n, at distance h from its unique strong witness, and a single source $\nu \in \mathcal{Q}_{s,D}$ for which, simultaneously for every clause selector,*

$$\tau \geq c_{k,r}\Psi_s(h + \log D + 1) \qquad \nu\text{-almost surely.}$$

The number of variables satisfies $n = O_{k,r}(\Psi_s(h + \log D + 1))$. In particular, (10) is sharp, uniformly in initial distance, source slack, and $s \in [s_*, 1]$.

One source forces the entire delay, for every selector: at the threshold and fixed D it has order h^3 against an independent-source expectation of $O_{k,r}(h)$. With $\delta = s - s_* > 0$ the crossover is at $\delta L^2 \asymp 1$, the cubic bound governing smaller L and L/δ larger L.

Why the delay is forced with probability one. By Theorem 2.1, the worst-case probability of $\{\tau > T\}$ is one exactly when this event has capacity at least $1/D$, and tree max-flow then supplies a source supported entirely on executions that survive time T. For the complete formulas, the capacity is the cover recursion on the distance line. For $T = 0$ every live distance has value one. As T grows, the values first drop near the end adjacent to the witness, and the frontier of the region where the value is still exactly one moves away from that end at speed at most of order $\epsilon + m^{-2}$ when it is at distance m, where $\epsilon = (s - s_*)\log k$ (Lemma C.3). The initial distance stays inside the value-one region until the frontier reaches it, which takes cubic time at criticality and linear time at a fixed margin; this handles $D = 1$. The cube-root-to-linear transition of such a frontier is known in branching processes [Mal19, BLMS24]. Appendix C proves the comparison and gives the separate cover lower bound that supplies the dependence on D.

An observable restart rule. The cover proof identifies excursions far from a strong witness as the obstruction to a better tail. Although the witness is unknown, a radius bound lets us cut off these excursions using only the current and initial assignments.

Theorem 4.4 (Restarting on leaving a Hamming ball)**.** *Suppose the formula has a strong witness x^* at Hamming distance $h \geq 1$ from the initial assignment x_0. Let $H \geq h$ be a supplied positive integer. Run the violated-clause walk, and reset to x_0 whenever an unsatisfying assignment leaves the Hamming ball of radius $2H$ around x_0. Continue with the next unused source symbol. The rule uses H and x_0, but does not use x^*. There are constants $c, C > 0$, depending only on k, r, such that for every clause selector, $D \geq 1$, $s \in [s_*, 1]$, and $\nu \in \mathcal{Q}_{s,D}$, its stopping time τ_H satisfies*

$$\mathbb{P}_\nu\{\tau_H > T\} \leq \min\{1, CD\lambda^h e^{-cT/H^2}\}, \qquad T \in \mathbb{Z}_{\geq 0}. \tag{13}$$

In particular,

$$\mathbb{E}_\nu \tau_H = O_{k,r}\Big(H^2(h + \log D + 1)\Big).$$

Here τ_H counts source-driven flips; including resets, the total number of variable updates is at most twice that, the distance from x_0 is maintained in constant work per flip, and work to find a violated clause is separate.

At $s = s_*$, a failure probability at most ε costs $O_{k,r}(H^2(h + \log D + \log(1/\varepsilon) + 1))$ flips, against $O_{k,r}((h + \log D + \log(1/\varepsilon) + 1)^3)$ for the original walk: for $H \asymp h$ and fixed D both are cubic in h, and the gain is that the confidence enters linearly rather than cubically. The algorithm needs neither the witness nor its distance h, only a valid upper bound H; $H = n$ is always valid but triggers no resets, and a smaller bound gives the tail scale H^2 independently of n.

The rule works because an assignment y just outside the ball has $\operatorname{dist}(y, x^*) \geq 2H + 1 - h > h$ by the triangle inequality, so resetting to x_0 strictly decreases the unknown witness distance; the restarted walk keeps that distance at most $h + 2H$ and retains the favorable-flip comparison, and a single-depth estimate on this bounded range, through the largest eigenvalue of the weighted transitions, gives the exponential tail (Appendix C). The barrier analysis thus yields an observable change to the algorithm, which then admits a simpler analysis.

The supplied radius can be removed by replaying the archived tape in copies with geometrically increasing radii, at the price of $O(\log(n + 2))$ search states and tape storage (Theorem C.4).

5 Noisy predecessor search under a prefix probability bound

Noisy search must both finish and return the right answer. Under independent errors, repeated checks can provide confidence. Under our source condition, consecutive errors may be predictable from the past, so the checks must be analyzed through the whole sequence of errors so far rather than one check at a time. We use verified backtracking and verification chains, which are classical [FRPU94, Section 3]. Our chain lengths depend on leaf depth, giving a guarantee in terms of the correct leaf's depth without knowing it or the tree height; sharper bounds under independent noise are known [EGW19, DLU25].

Let $x_1 < \cdots < x_n$ $(n \geq 1)$ be sorted keys and let $x \in \mathbb{R}$ be the target. The task is to return the index of the largest key not exceeding x, the *predecessor* of x. A primitive query asks whether $x < x_i$. Write $e_t = 1$ when the answer to the tth primitive query is flipped and $e_t = 0$ otherwise. Fix $1/2 < p < 1$ and assume, for every prefix e of the error sequence (e_t),

$$\nu[e] \leq D\big(p^{|e|-|e|_1}(1-p)^{|e|_1}\big)^s. \tag{14}$$

Thus the reference law P has independent errors of probability $1 - p$; the actual law need not give any conditional correctness guarantee.

The walk on the tree. Fix an ordered binary search tree whose leaves represent the predecessor intervals, so that an internal node represents an interval $[x_i, x_j)$ and splits it at some key. The algorithm walks on this tree. At each node it makes three logical comparisons. Two *verification* comparisons test whether x lies in the node's interval, by asking whether $x < x_i$ and whether $x < x_j$; one *split* comparison chooses the child. If verification fails, the walk moves to the parent; otherwise it descends to the chosen child. This is verified backtracking [FRPU94].

Each original leaf is extended by a chain of nodes that all represent the leaf's interval, and a chain node's parent is the previous node of the chain. At a chain node the split comparison is replaced by a dummy comparison, a primitive query that is made but whose answer is ignored; verification continues, failed verification moves back one node, and success advances along the chain. The walk halts at the first chain endpoint it reaches and returns that leaf's label (comparisons against an infinite boundary are likewise dummies). Every logical comparison is answered by the majority of ℓ repetitions of the same primitive query, ℓ odd and fixed below, and all 3ℓ primitive queries at a node are made before the move, whether or not their answers are needed; so every move consumes exactly 3ℓ raw error symbols, and the source condition applies block by block.

The chains serve the error guarantee. Measure distance from the endpoint of the correct leaf's chain: a move with all three majorities correct brings the walk one step closer to it, by correctly rejecting a node whose interval excludes x or correctly descending or advancing from one whose interval contains it, and any other move changes the distance by at most one. Longer chains at deeper leaves place every wrong endpoint farther from the correct endpoint than the walk's starting point, by a margin that controls all incorrect outputs at once.

The source threshold. Choose the least odd $\ell = \ell(p)$ for which the majority of ℓ independent reference answers is correct with probability greater than $2^{-1/3}$, so that three independent majorities are all correct with probability greater than $1/2$. Let $G \subseteq \{0,1\}^{3\ell}$ contain the raw error blocks in which all three majorities are correct, and put

$$a_s = \sum_{e \in G} P[e]^s, \qquad b_s = \sum_{e \in \{0,1\}^{3\ell} \setminus G} P[e]^s, \qquad 2\sqrt{a_{s_c} b_{s_c}} = 1, \qquad \lambda = \sqrt{a_{s_c}/b_{s_c}}.$$

These play the roles of a_s, b_s, and λ in Section 4, with a block of 3ℓ error symbols in place of one label. The solution $s_c \in (0,1)$ exists and is unique because $a_s b_s$ is strictly decreasing in s, and $\lambda > 1$ because $a_{s_c} > 1/2$ (Appendix F). Blocks in G are the favorable moves above; every other block is treated as potentially adverse, with no independence assumed among the majority answers, so s_c is a sufficient threshold rather than a sharp one.

Theorem 5.1 (Verified search with correlated errors)**.** *Take any ordered binary search tree whose leaves represent the predecessor intervals, and let h be the depth of the correct leaf. Put $L_\eta = \max\{1, \lceil \log_\lambda(D/\eta) \rceil\}$ and attach a chain of $\mathrm{depth}(u) + L_\eta$ edges to each original leaf u. For every $D \geq 1$, $0 < \eta < 1$, $s \in [s_c, 1]$, and law satisfying (14), the walk above halts almost surely and returns the predecessor of x with error probability at most η. The number Q of primitive queries satisfies*

$$\mathbb{E}_\nu Q \leq C_p \min\left\{K^3, \frac{K}{s - s_c}\right\}, \qquad K = h + \log D + \log(1/\eta) + 1,$$

where $K/0 = \infty$. The algorithm uses p, D, η, but needs neither h nor a bound on the tree height; chains are implemented by counters. The exact query schedule is specified in Appendix F.

The running-time bound is the barrier principle of Section 4 applied to the distance from the correct endpoint, which starts at $2h + L_\eta$. The error bound uses one cover: before any incorrect

output, the walk must pass a live state whose distance from the correct endpoint exceeds its starting distance by more than L_η, and the harmonic first-hit bound prices that excursion at $D\lambda^{-L_\eta-1} \le \eta$, with no union bound over leaves. The correct endpoint is used only in the proof. For a balanced tree $h = O(\log(n+1))$; on an unbalanced tree the guarantee depends on the queried leaf's depth and requires no prior bound on it.

6 When restarting restores exponential tails

A restart changes the algorithm's state, but does not replenish the randomness in the tape. A prefix cover nevertheless tells us when to abandon an attempt: restart when the observed word reaches the cover. The criterion below shows that robust almost-sure termination and an exponential tail are possible under exactly the same strict cover inequality, and that nothing can be done when it fails.

Fix an initial configuration x and a deterministic algorithm that consumes one source symbol per step. A restart resets the algorithm to x and resumes with the next unused symbol. The policy is deterministic, uses only observed symbols, and may restart at most once between successive steps. Running time counts source-consuming steps; reset costs are separate. Write $\mathcal{N}_x$ for the original nontermination event and $\alpha_s(x) = \mathcal{C}_s(\mathcal{N}_x)$, where $0 < s \le 1$.

Theorem 6.1 (Exact restart criterion)**.** *The following statements are equivalent:*

(i) $\alpha_s(x) < 1$.

(ii) Some restart policy and constants $K < \infty$, $0 < \gamma < 1$ *satisfy* $\Pr_\nu\{\tau_{\text{restart}} > t\} \le DK\gamma^t$ *for every* $D \ge 1$, $\nu \in \mathcal{Q}_{s,D}$, *and integer* $t \ge 0$.

(iii) Some restart policy terminates almost surely for every $\nu \in \mathcal{Q}_{s,1}$.

If $\alpha_s(x) = 1$, *every restart policy has nontermination probability one under some source in* $\mathcal{Q}_{s,1}$.

Proof of sufficiency. If $\alpha_s(x) < 1$, the continuity (4) of finite-horizon capacities gives a horizon T and a finite prefix-free cover $\mathcal{B}$ of $\{\tau > T\}$, with words of length at most T, whose cost $c = \sum_{b\in\mathcal{B}} P[b]^s$ is less than one. Remove words on which termination has already occurred. Restart on reaching a word in $\mathcal{B}$. Every attempt ends within T steps. The words describing j failed attempts form the prefix-free family $\mathcal{B}^j$, of total cost c^j. Consequently,

$$\Pr_\nu\{\tau_{\text{restart}} > jT\} \le D \sum_{w\in\mathcal{B}^j} P[w]^s = Dc^j.$$

This applies the prefix bound to the execution as a whole, across attempts; it requires no conditional entropy guarantee after a restart. The converse is proved in Appendix D. □

For the converse, if $\alpha_s(x) = 1$ the capacity identity gives a source μ with $D = 1$ supported on nontermination, and against any restart policy the adversary restarts μ along with the algorithm: the concatenation of independent copies of μ, one per attempt, still satisfies the prefix bound, and every attempt follows a nonterminating execution until the policy abandons it.

Synthesis for explicit finite-state processes. For a finite explicit state space, one can synthesize the updates as well as the restart rule; the problem is minimax control against a source constrained along the whole trajectory rather than at each state. Let X be the live states, let F_y be a finite nonempty set of available updates at y, and let $\mathcal{T}_f(y,a)$ be the state reached when update f reads symbol a. Set

$$R_f(y,z) = \sum_{a:\mathcal{T}_f(y,a)=z} \pi(a)^s, \qquad y,z \in X,$$

omitting terminated states; $R_f(y,\cdot)$ is the row of weights that update f contributes to the cover recursion at y. Updates are selected before reading the next symbol; selectors have no additional private randomness. In the linear program below, the constraints say that v is a subsolution of the cover recursion under every update, $v(y)$ at most its one-step cost for each f, and maximizing picks the largest subsolution, which is the infinite-horizon value: $V(y)$ is the worst-case nontermination capacity from y under the best selector.

Theorem 6.2 (Synthesis of robust selection and restarts)**.** *Let V be the unique optimizer of*

$$\max\left\{\sum_{y\in X} v(y) : 0 \le v \le \mathbf{1}, \quad v(y) \le \sum_{z\in X} R_f(y,z)v(z) \text{ for every } y\in X,\ f\in F_y\right\}. \tag{15}$$

(i) (Optimal selection.) *There is a deterministic stationary selector μ such that, for every initial state x and every $D \ge 1$,*

$$\max_{\nu\in\mathcal{Q}_{s,D}} \mathbb{P}_\nu^{\mu,x}\{\tau=\infty\} = \min_{\Lambda}\max_{\nu\in\mathcal{Q}_{s,D}} \mathbb{P}_\nu^{\Lambda,x}\{\tau=\infty\} = \min\{1, DV(x)\},$$

where Λ ranges over deterministic nonanticipating selectors without restarts.

(ii) (Restarting.) *Fix a reset state x_0. Some deterministic nonanticipating policy, allowed to select updates and restart to x_0, terminates almost surely for every source in $\mathcal{Q}_{s,1}$ if and only if $V(x_0) < 1$. When $V(x_0) < 1$, using μ and restarting on entry to $S = \{y : V(y) = 1\}$ gives constants $K < \infty$ and $0 < \gamma < 1$ such that $\mathbb{P}_\nu\{\tau_{\text{restart}} > t\} \le DK\gamma^t$ for every $D \ge 1$, $\nu \in \mathcal{Q}_{s,D}$, and integer $t \ge 0$. When $V(x_0) = 1$, every such policy has nontermination probability one under some source in $\mathcal{Q}_{s,1}$.*

(iii) (Computation.) *If the powered coefficients $R_f(y,z)$ are rational, then V, μ, the restart set S, and explicit constants for the tail bound are computable in polynomial time in the explicit state–action description.*

The linear program (15) is the classical one of affine monotone dynamic programming [Ber19]; here it yields exact guarantees against globally constrained tapes, including an impossibility result for all history-dependent policies.

The first linear program determines the optimal values, but an update that merely satisfies the cover recursion need not attain them: at a state with $V = 0$, an update that always terminates and one that loops forever both satisfy it. A second linear program resolves these ties on the states with $V < 1$. It produces a positive potential ψ with $\psi = \mathbf{1} + A\psi$, where A is the selected powered transition matrix on those states and $\mathbf{1}$ is the all-ones vector. Thus $A\psi < \psi$, which forces geometric decay, and an adjusted potential proves decay after transitions into $\{V = 1\}$ are redirected to the reset state. The proof in Appendix D again sums powered probabilities of prefixes of the whole execution tape. An implicit state space can be exponentially large; the Hamming-ball rule of Theorem 4.4 avoids the enumeration by using the geometry of the assignments.

Adaptive restarts can beat every fixed schedule. Classical restart schedules, such as Luby, Sinclair, and Zuckerman's universal schedule [LSZ93], choose cutoff times in advance; dynamic policies and models of dependence among runs are studied in [KHR+02, RHK02]. Under a shared tape, the time to abandon an attempt may need to depend on what has been observed, and no predetermined schedule can always substitute for an adaptive rule: Proposition D.1 in Appendix D gives a three-state procedure with reference mean running time 8/3 for which every predetermined

cutoff sequence has worst-case nontermination probability at least $1/2$ in $\mathcal{Q}_{1/2,1}$, whereas one stationary rule has exact worst-case tail $\min\{1, DT2^{1-T}\}$: the procedure has a state in which the source can cheaply hold the walk for any prescribed time, a predetermined cutoff tells the adversary how long that is, and the rule that restarts immediately on entering that state never gives it the chance.

7 Conclusion and further questions

The independent-source stopping-time law does not determine worst-case behavior under a prefix probability bound. Covers of varying depth recover what the law loses, exactly, and the recovered information is algorithmic: it selects the order in which to attempt tasks, the source threshold of a local walk, and the moment to restart.

Within the model, four questions remain: whether the noisy-search threshold s_c is sharp; whether a single SAT walk can match the replay guarantee without storing the tape; whether compact state representations can replace the explicit enumeration of Theorem 6.2, as the Hamming ball does in Theorem 4.4; and whether there are natural algorithms for which covers using any fixed number of depths are asymptotically weaker than general covers. Beyond it, the synthesis theorem assumes that the reference law and the transition map are known, and every analysis assumes a fixed reference law. When the model must be learned, what additional assumptions allow a policy that reaches a target reliably under a prefix probability bound on the inputs? And when proposal distributions adapt to the observed history, as in search with externally verified candidates, can covers still guide branching and restarting?

A Detailed related work

Our source class belongs to the study of computation with weak randomness. The Chor–Goldreich sources [CG88] bound each block's min-entropy conditionally on all previous blocks; the almost Chor–Goldreich sources of Doron, Moshkovitz, Oh, and Zuckerman [DMOZ23] let a bounded number of block positions, fixed in advance, be arbitrary and the remaining blocks be close to the required conditional min-entropy, and are used there to drive random walks on expanders that accumulate entropy and yield online condensers. Our condition is cumulative rather than positional: which symbols are predictable may depend on the history, as long as every prefix stays within its budget.

The same authors [DMOZ25, Definition 2] study *unpredictable sources*: averaged over positions and draws of the source, the realized next symbol has large conditional probability only rarely. Their model permits predictable steps and includes arbitrary finite sources of sufficiently high min-entropy rate, for suitable block parameters. They analyze entropy accumulation on lossless expanders and construct online condensers. Our source class overlaps with this setting. For a uniform binary reference it requires $H_\infty(X_{1:t}) \geq st - \log_2 D$ at every length t, so finite truncations have high min-entropy rate when s is close to one and t is sufficiently large. A bound at one terminal horizon need not constrain short prefixes this way. We determine exact survival probabilities and use them to choose retry orders and restarts on the same tape. Related work on Markov-chain robustness controls changes to individual transition probabilities [Zuc26].

The source-capacity identity is classical tree max-flow/min-cut [BP17]; its dimension interpretation draws on connections among symbolic spaces, entropy, and Hausdorff dimension [Sta93, MW88, Sim15]. Lutz [Lut11] studies constructive and finite-state versions of Billingsley dimension, which measure dimension relative to a reference law. Our covers and source laws carry no computability restriction. Achlioptas, Iliopoulos, and Vlassis [AIV17, Section 4] cut each trajectory when its probability crosses a threshold, using prefixes of different lengths in their entropy-compression analysis. The capped recurrence has the familiar multiplicative optimal-stopping form [JPS21], and the finite-state synthesis uses affine monotone dynamic programming [Ber19]. In our setting the capacity identity (3) gives exact source probabilities and guides updates and restarts.

Two perspectives from learning theory motivated this formulation: the hierarchy of local descriptions behind the pointwise generalization bounds of Li and Xu [LX26, Section 3.3], and the question, formalized in [Xu26, Definition 2.2 and Proposition 2.3], of which history summaries preserve what a Bellman recursion needs. The analogous question in this paper is which execution summaries suffice for control; separate summaries at every depth can lose the way prefixes extend one another. The connection is conceptual; our proofs use the prefix-source constraint and tree capacity.

The promise-SAT walk and its distance analysis are due to Austrin, Guruswami, and Håstad [AGH17, Proposition 6.1]. Search within Hamming balls is also established [DHW03]; here repeated resets return to the same center under one cumulative tape constraint. The noisy-search precedents are discussed in Section 5.

The cubic scale and its crossover have precedents in probability, statistical physics, and algorithmic search. The asymptotic analysis of Derrida and Simon [DS07] predicts critical survival of order $\exp[-\Theta(t^{1/3})]$ from fixed positive distance for branching with an absorbing wall, with a near-critical crossover governed by approximate boundary motion of the form $m'(t) = \epsilon + c/m(t)^2$. Lemma C.3 proves the probability-one plateau directly, in the discrete capped form with the near-critical slack that the source construction needs; at criticality, Mallein's consistent-displacement theorem [Mal19, Theorem 1.1] also implies it, as explained after that lemma. Berestycki, Liu, Mallein, and Schweinsberg [BLMS24] prove a related transition between cube-root and linear displacement at time of order $\epsilon^{-3/2}$ for branching Brownian motion. Killed-branching estimates also enter computational

complexity: Pemantle [Pem09] uses them to bound the cost of finding nearly optimal paths in random trees. Our contribution is the sharp source-robust characterization of the specified SAT walk. The upper bound allows arbitrary clause overlap and history-dependent selection; the matching construction gives one admissible source forcing the delay with probability one for every selector. The proof establishes exact saturation of the discrete capped recursion and uses the capacity identity to construct the source.

B Prefix covers and worst-case sources

Throughout this appendix, $P = \pi^{\mathbb{N}}$ is the full-support finite source from Section 2. The source class $\mathcal{Q}_{s,D}$ imposes its bound on every prefix of the full tape. Symbols after termination may be continued with the reference law.

Remarks on the source class. Replacing the repeated block of the introductory stream by m fixed symbols preserves the prefix bound and makes the coordinates mutually independent, so a lack of fresh randomness can arise with or without dependence. Independent bits with bias p satisfy the fair-bit version of the condition whenever $\max\{p, 1-p\} \le 2^{-s}$. The bound is information-theoretic, not a test for pseudorandomness: a generator with at most 2^b seeds has, at every length, some output prefix of probability at least 2^{-b}, so a fixed finite seed cannot satisfy the condition forever. This limits which sources the theorems cover; it is not a lower bound on the randomness needed by every SAT solver, and finite-time conclusions require the prefix inequalities only through the deadline in question.

Proof of Theorem 2.1. Every prefix cover gives $\nu(E) \le D \sum_{w \in \mathcal{B}} P[w]^s$, which proves the upper bound. Suppose first that E is determined by T symbols. In the full source tree through depth T, give the root capacity $c(\varnothing) = 1$ and every other node w capacity $c(w) = DP[w]^s$. Route flow from the root only into the depth-T words whose cylinders lie in E. The tree max-flow/min-cut theorem gives a largest admissible flow f of value $\min\{1, D\mathcal{C}_s(E)\}$. Indeed, cuts are exactly prefix covers, with root cost 1; a cover of a whole depth-T cylinder by longer words cannot lower its cost, since for nonnegative x_i and $s \le 1$, $\sum_i x_i^s \ge (\sum_i x_i)^s$.

It remains to extend f to a probability law on all depth-T words. Extend f by zero on branches outside its support. For every internal node, including the root,

$$\sum_a c(wa) \ge c(w).$$

For nonroot nodes this follows from $\sum_a \pi(a)^s \ge 1$; at the root it also uses $D \ge 1$. Since f conserves flow, the residual capacities $r(w) = c(w) - f(w)$ satisfy

$$\sum_a r(wa) \ge r(w).$$

Start an additional flow g of value $1 - f(\varnothing)$ at the root. Whenever its incoming mass at w is at most $r(w)$, distribute that mass among its children without exceeding their residual capacities. The displayed inequality makes this possible at every step. Then $f + g$ has total mass one and respects all node capacities. It cannot give E more than the maximal value of f, because its restriction to successful leaves would itself be a larger admissible flow. Thus this probability law attains the upper bound. Continue it beyond depth T using the reference conditional law. For any suffix v,

$$\nu[wv] = \nu[w]P[v] \le DP[w]^s P[v]^s = DP[wv]^s,$$

so it belongs to $\mathcal{Q}_{s,D}$ on the infinite tape space.

For a closed E, let E_n be the union of the length-n cylinders that intersect E. These are decreasing closed-and-open sets with intersection E. Their contents decrease to $\mathcal{C}_s(E)$: monotonicity gives one inequality, while any finite prefix cover of E covers all sufficiently large E_n, by compactness. The class $\mathcal{Q}_{s,D}$ is nonempty and weakly compact. Indeed, $P \in \mathcal{Q}_{s,D}$, the tape space is compact, and each cylinder constraint is closed under weak convergence. Choose maximizers ν_n for E_n and a weakly convergent subsequence with limit $\nu_* \in \mathcal{Q}_{s,D}$. For fixed m and $n \geq m$,

$$\nu_n(E_m) \geq \nu_n(E_n) = \min\{1, D\mathcal{C}_s(E_n)\}.$$

Pass to the subsequential limit, using that E_m is closed and open, and then let $m \to \infty$. Continuity from above yields $\nu_*(E) \geq \min\{1, D\mathcal{C}_s(E)\}$, proving attainment and (3). For any decreasing sequence of finite-symbol events E_T with intersection $\mathcal{N}$, the same compactness argument shows $\mathcal{C}_s(E_T) \downarrow \mathcal{C}_s(\mathcal{N})$ and proves (4). □

Corollary B.1 (Dimension as a source threshold)**.** *Let $\mathcal{N}$ be closed and put $\delta = \dim_H^P \mathcal{N}$. For $0 < s < \delta$, some finite D and some $\nu \in \mathcal{Q}_{s,D}$ satisfy $\nu(\mathcal{N}) = 1$. For $\delta < s \leq 1$, every such law gives $\nu(\mathcal{N}) = 0$. For every $0 < s \leq 1$, the least domination constant permitting a law supported on $\mathcal{N}$ is*

$$\inf\{D \geq 1 : \exists \nu \in \mathcal{Q}_{s,D},\ \nu(\mathcal{N}) = 1\} = \frac{1}{\mathcal{C}_s(\mathcal{N})}, \tag{16}$$

where $1/0 = \infty$. If $P(\mathcal{N}) = 0$, then

$$\delta = \inf\{s \in (0, 1] : \nu(\mathcal{N}) = 0 \text{ for every finite } D \geq 1 \text{ and every } \nu \in \mathcal{Q}_{s,D}\}. \tag{17}$$

At $s = \delta > 0$, there exists a law with finite domination constant supported on $\mathcal{N}$ exactly when $\mathcal{C}_\delta(\mathcal{N}) > 0$; dimension alone does not settle this endpoint.

Proof of Corollary B.1. In the metric $d_P(\omega, \omega') = P[\omega \wedge \omega']$, each cylinder $[w]$ has diameter $P[w]$. Any nonsingleton set is contained in a cylinder with the same diameter; singleton sets can be covered by cylinders of arbitrarily small diameter. Thus the infimum over countable cylinder covers equals the Hausdorff content. For compact $\mathcal{N}$, every such cover has a finite subcover, so this infimum is $\mathcal{C}_s(\mathcal{N})$. Its zero threshold is therefore $\dim_H^P \mathcal{N}$: the content is positive for $s < \delta$ and zero for $s > \delta$. Theorem 2.1 proves (16) and the strict-threshold claims. If $P(\mathcal{N}) = 0$, then $\mathcal{C}_1(\mathcal{N}) = P(\mathcal{N}) = 0$, so the set in (17) is nonempty and has infimum δ. The same exact identity gives the stated critical criterion. □

B.1 Controlled prefix capacities

For $0 < s \leq 1$, take a finite state space, a common finite symbol alphabet with full-support probabilities $\pi(a)$, and a finite nonempty set F_x of available updates at each live state. Write $P = \pi^{\mathbb{N}}$ and let $\mathcal{T}_f(x, a)$ be the state obtained when update f uses symbol a at state x. Set $V_0(x) = 1$ at live states and $V_t(x) = 0$ at terminated states. At live states define

$$V_{t+1}(x) = \min\left\{1, \min_{f \in F_x} \sum_a \pi(a)^s V_t(\mathcal{T}_f(x, a))\right\}. \tag{18}$$

Proposition B.2 (Optimal deterministic selection)**.** *For every $D \geq 1$ and finite horizon T,*

$$\min_{\Lambda} \max_{\nu \in \mathcal{Q}_{s,D}} \mathbb{P}_{\nu}^{\Lambda}\{\tau > T\} = \min\{1, DV_T(x_0)\}, \tag{19}$$

where Λ ranges over deterministic nonanticipating selectors. One selector depending only on current state and remaining time is optimal simultaneously for every $D \geq 1$ and every initial state.

This recursion evaluates a single globally constrained source; it does not impose independent conditional constraints after successive symbols.

Proof of Proposition B.2. Fix a finite horizon. At a live root, a prefix cover either cuts the root at cost one or covers the children after the chosen update. For a specified first update f, the selector can choose its continuation independently after each observed symbol a. The optimal continuation cost is $V_t(\mathcal{T}_f(x,a))$, multiplied by the edge cost $\pi(a)^s$. Minimizing over f and comparing with the root cut gives (18) by induction. At the horizon, the surviving event costs one at a live state and zero at a terminated state, exactly the stated boundary values. Each minimum is attained, and an attaining choice can depend only on current state and remaining time. These choices depend on s but not on D, and the same state-based rule is available from every initial state. Theorem 2.1 now turns the optimal cover cost into (19), since $c \mapsto \min\{1, Dc\}$ is nondecreasing.

This proof optimizes one globally constrained source. It does not reset the domination constant after a prefix or give the source independent conditional choices at each step. The selector has no additional private randomness; allowing trusted private coins would define a different optimization problem. □

C Proofs for the promise-SAT walk

C.1 A barrier bound from an integer potential

The proof has two parts. A harmonic function pays for paths when they first reach a distant barrier. A spectral estimate pays for paths that remain below it. Both bounds are on prefix weights; neither treats the actual source as a Markov chain.

The upper bound uses only a property of the potential after each possible source symbol. In particular, it needs neither a finite state space nor a Markov transition law for the potential.

Lemma C.1 (Integer-potential barrier)**.** *Fix $k/2 < r < k$ and the parameters λ, s_* in (8). Consider a deterministic process driven by symbols in $[k]$, with uniform product reference law P and stopping time τ. Assign to every live history a positive integer potential d, initially $h \geq 1$, and give every terminated history potential zero. At each live history, assume that at least r of the k next symbols either terminate or give a live potential at most $d-1$, and that every other symbol either terminates or gives a live potential at most $d+1$. For every integer $m > h$, every $T \geq 0$, every $D \geq 1$, and every $\nu \in \mathcal{Q}_{s_*,D}$,*

$$\mathbb{P}_{\nu}\{\tau > T\} \leq \min\left\{1, D\left[\frac{h}{m}\lambda^{h-m} + \frac{\lambda^h}{\sqrt{\lambda^2-1}}\exp\left(-\frac{\pi^2 T}{2(2m-1)^2}\right)\right]\right\}. \tag{20}$$

Consequently the critical tail and cubic expectation bounds in Theorem 4.1 hold.

Proof. Put

$$a = rk^{-s_*} = \lambda/2, \qquad b = (k-r)k^{-s_*} = 1/(2\lambda).$$

If f is nonnegative and increasing, with $f(0) = 0$, the sum of f over the live children of a history of potential d, multiplied by the powered edge weight k^{-s_*}, is at most

$$af(d-1) + bf(d+1). \tag{21}$$

Indeed, assign r of the decreasing symbols to the first term. Each other live child has potential at most $d+1$; terminated children contribute zero. This argument applies separately after every history.

Cover $\{\tau > T\}$ by prefixes that first reach potential m while still live, up to time T, and live length-T prefixes that never reach m. Remove any length-T prefix already in the first set. Because upward increments are at most one, the first set meets every path that leaves $\{1, \dots, m-1\}$. The resulting cover is prefix-free.

For first-hit prefixes use

$$H_m(d) = \frac{d}{m}\lambda^{d-m} \qquad (0 \le d \le m).$$

This function is increasing, $H_m(0) = 0$, $H_m(m) = 1$, and

$$aH_m(d-1) + bH_m(d+1) = H_m(d) \qquad (1 \le d < m).$$

Start with weight $H_m(h)$ at the root. Expanding a live node below m cannot increase the sum of its powered prefix weight times H_m, by (21). Leave first-hit nodes unexpanded, give terminated nodes value zero, and continue to depth T. Every first-hit node has value one; the remaining values are nonnegative. Finite-tree induction therefore bounds the weight of first-hit prefixes by $H_m(h)$.

For paths remaining below m, put $L = m-1$ and define a reflected comparison matrix $R^{(m)}$ on $\{1, \dots, L\}$ by

$$(R^{(m)}f)(d) = af(d-1) + bf(d+1) \quad (d < L), \qquad (R^{(m)}f)(L) = af(L-1) + bf(L),$$

where $f(0) = 0$. This matrix preserves nonnegative increasing vectors: consecutive differences in the interior are nonnegative linear combinations of consecutive differences of f, and the final difference is $a(f(L-1) - f(L-2)) \ge 0$ when $L \ge 2$. The claim is immediate for $L = 1$. Thus $F_t = (R^{(m)})^t\mathbf{1}$ is nonnegative and increasing. Induction on the remaining horizon bounds the weight of length-T live paths that stay below m by $F_T(h)$. For a history with potential below L, apply (21) to F_t. At potential L, an upward exit is discarded, each of r decreasing symbols contributes at most $F_t(L-1)$, and any other retained child contributes at most $F_t(L)$. This gives the reflected final row.

Let $B = \mathrm{diag}(\lambda, \dots, \lambda^L)$. The symmetric matrix $B^{-1}R^{(m)}B$ has off-diagonal entries $1/2$, last diagonal entry $b < 1/2$, and other diagonal entries zero. Increasing the last diagonal entry to $1/2$ gives a nonnegative symmetric matrix with largest eigenvalue $\cos(\pi/(2L+1))$. To see the latter, the positive vector $v_j = \sin(j\pi/(2L+1))$ satisfies its eigenvector equation, including the reflected final row. A symmetric nonnegative matrix has Euclidean operator norm equal to its spectral radius. Perron–Frobenius monotonicity consequently gives

$$\|B^{-1}R^{(m)}B\|_2 \le \cos\frac{\pi}{2m-1}.$$

Since $\|(\lambda^{-j})_{j=1}^{L}\|_2 \le 1/\sqrt{\lambda^2-1}$,

$$F_T(h) \le \frac{\lambda^h}{\sqrt{\lambda^2-1}}\left(\cos\frac{\pi}{2m-1}\right)^T \le \frac{\lambda^h}{\sqrt{\lambda^2-1}}\exp\left(-\frac{\pi^2 T}{2(2m-1)^2}\right).$$

The source constraint bounds the probability of each cover cylinder by D times its powered reference mass. Summing the two parts proves (20).

When $T^{1/3} > h$, choose an integer $m > h$ within two of $T^{1/3}$. Both terms are at most $C\lambda^h e^{-cT^{1/3}}$. When $T^{1/3} \le h+1$, the same capped bound follows from probability at most one, after enlarging C and decreasing c. Finally, summing $\min\{1, CD\lambda^h e^{-cT^{1/3}}\}$ over integer $T \ge 0$ gives $C'(h+\log D+1)^3$. For example, split the sum at a sufficiently large constant times $(h+\log D+1)^3$; beyond that point its summands are bounded by $e^{-cT^{1/3}/2}$. □

Proof of Theorem 4.1. Let d be the Hamming distance to x^*. Whenever a clause is violated, every literal it satisfies under x^* belongs to a coordinate on which the current assignment and x^* disagree. At least r flips therefore decrease d by one; all others increase it by one. Assign potential zero on termination. Every live assignment has positive potential. The barrier lemma and $\mathcal{Q}_{s,D} \subseteq \mathcal{Q}_{s_*,D}$ give the critical tail and the cubic expectation bound. An initially satisfying assignment has $\tau = 0$.

For the additional bound above the threshold, write $\delta = s - s_*$ and sum over live length-t histories:

$$M_t = \sum_{\substack{|w|=t \\ \tau(w)>t}} k^{-st}\lambda^{d(w)}.$$

At each live history the sum of the children's potential weights is at most

$$k^{-s}\big(r\lambda^{d-1} + (k-r)\lambda^{d+1}\big) = k^{-\delta}\lambda^d.$$

Terminated children contribute zero. Thus $M_T \le \lambda^h k^{-\delta T}$ by finite-tree induction. Every live leaf has $\lambda^{d(w)} \ge 1$, so the prefix bound gives $\mathbb{P}_\nu\{\tau > T\} \le DM_T$. For $\delta > 0$, summing the capped exponential tail gives $O_{k,r}(L/\delta)$, where $L = h + \log D + 1$. Combining this with the cubic bound proves (10). This weighted-tree argument also applies to the general potential in Lemma C.1; it uses no conditional probability assumption on the source. Under P, the expected decrease in distance at a live history is at least $(2r-k)/k$, giving $\mathbb{E}_P\tau \le kh/(2r-k)$ by additive drift. □

The single-depth baseline. For SAT, live distances are at most n. Use the reflected matrix above on $\{1,\dots,n\}$. At distance n, every actual flip decreases distance, so its powered contribution on an increasing vector f is at most $(a+b)f(n-1) \le af(n-1) + bf(n)$. The same induction and spectral estimate therefore bound the single-depth live-prefix sum by

$$\widetilde{Z}_T(s_*) := \sum_{w\in[k]^T:\tau(w)>T} k^{-s_*T} \le \frac{\lambda^h}{\sqrt{\lambda^2-1}} \exp\left(-\frac{\pi^2 T}{2(2n+1)^2}\right). \tag{22}$$

Applying the source bound at depth T and summing the capped tail proves $\mathbb{E}_\nu\tau = O_{k,r}(n^2(h + \log D + 1))$.

C.2 A family with a sharp source threshold

For $n \ge k+1$, let Φ_n contain every clause

$$\left(\bigvee_{i\in I} x_i\right) \vee \left(\bigvee_{j\in J} \neg x_j\right), \quad I \cap J = \varnothing, \quad |I| = r, \quad |J| = k-r. \tag{23}$$

There are $\binom{n}{k}\binom{k}{r} = O_k(n^k)$ clauses. The all-one assignment strongly r-satisfies the formula. It is the unique such assignment. For an assignment with $z \ge 1$ zeroes and $o = n - z$ ones, if $o \ge k - r$, put

$k - r$ ones in the negative positions and at least one zero in the positive positions; the clause has fewer than r true literals. If $o < k - r$, put all the ones in negative positions and fill the remaining positions with zeroes; the clause has $k - r - o < r$ true literals. The required choices exist because $n \geq k$.

Order positive literals first in each clause. If d is the number of zero variables, a clause is violated precisely when its positive variables are zero and its negative variables are one. Hence an assignment is live exactly when $r \leq d \leq n - k + r$. Put $N = n - k + 2$ and $j = d - r + 1$; the live values are $1, \ldots, N - 1$. For every selector, symbols $1, \ldots, r$ decrease j, and the other symbols increase it. The powered matrix of live transitions is

$$R_s(j, j-1) = rk^{-s}, \qquad R_s(j, j+1) = (k-r)k^{-s}, \tag{24}$$

with out-of-range entries omitted.

Proof of Theorem 4.2(i). The spectral radius of (24) is

$$\rho_s = 2\sqrt{r(k-r)}\, k^{-s} \cos(\pi/N).$$

For $s < s_*$, take N sufficiently large that $\rho_s > 1$. A positive right eigenvector is $\varphi(j) = \lambda^j \sin(j\pi/N)$. Start at a maximizing j_0, and at live value j give label a probability

$$q(a \mid j) = k^{-s} \frac{\varphi(j + \delta(a))}{\rho_s \varphi(j)}, \qquad \delta(a) = \begin{cases} -1, & a \leq r, \\ 1, & a > r, \end{cases}$$

where $\varphi(0) = \varphi(N) = 0$. The eigenvector equation normalizes these probabilities, and the boundary labels have probability zero. For every supported word w of length t, telescoping gives

$$\nu[w] = k^{-st} \rho_s^{-t} \frac{\varphi(j_t)}{\varphi(j_0)} \leq k^{-st} = P[w]^s.$$

The resulting source belongs to $\mathcal{Q}_{s,1}$ and never exits the live interval. The path j_t is determined by source labels alone, independently of the clause selector. Thus this one source forces nontermination for all selectors simultaneously. □

Corollary C.2 (Full support and nonperiodic failure)**.** *For every $0 < s < s_*$ and $0 < \varepsilon < 1$, there are a strongly r-satisfiable formula, a fixed initial assignment, and a single source $\nu \in \mathcal{Q}_{s,1}$ such that*

$$\nu\{\tau = \infty\} \geq 1 - \varepsilon$$

for every deterministic nonanticipating clause selector. The source can be generated by a time-homogeneous stochastic machine with finitely many internal states, satisfies $\nu[wa]/\nu[w] > 0$ for every finite word w and every symbol a, and almost surely produces a tape that is not eventually periodic. For each fixed deterministic memoryless clause selector, the assignment trajectory on nontermination is almost surely not eventually periodic.

The conditional probabilities need not have a common positive lower bound. The generator is stochastic, not a deterministic finite-seed PRNG. Nonperiodicity also does not mean that assignments are never revisited: a finite assignment space forces revisits on every infinite run. The statement rules out eventual repetition of a fixed cycle. The proof uses the source from Theorem 4.2(i).

Proof of Corollary C.2. Let $q \in \mathcal{Q}_{s,1}$ be the source in the proof of Theorem 4.2(i), and set $\nu = (1-\varepsilon)q + \varepsilon P$. Both component laws satisfy the same prefix bound, so ν does too. The q component forces nontermination for all selectors simultaneously, proving the probability bound. The generator for q has the finite internal state $j \in \{1, \dots, N-1\}$ and fixed transition probabilities; an initial choice of mode implements the mixture with P using only finitely many additional states. Furthermore, $\nu[wa] \geq \varepsilon k^{-|w|-1} > 0$.

Every individual tape has mass at most $\lim_{t\to\infty} k^{-st} = 0$. Since the eventually periodic tapes form a countable set, they have total mass zero. For a memoryless selector, the current assignment determines the selected clause, and the next assignment identifies the flipped coordinate and hence its unique literal index. Thus an eventually periodic assignment trajectory would determine an eventually periodic source tape. □

C.3 Matching tails and the gap from single-depth covers

Proof of Theorem 4.2(ii). Write $R = R_{s_*}$, $a = \lambda/2$, $b = 1/(2\lambda)$, and $\varepsilon = (s - s_*)\log k \geq 0$, so that $R_s = e^{-\varepsilon}R$. The exact distance reduction gives the prefix-capacity recursion

$$C_0 = \mathbf{1}, \qquad C_{t+1} = \min\{\mathbf{1}, e^{-\varepsilon}RC_t\},$$

with the minimum taken coordinatewise, and the prefix-cover identity gives $A_{N,T}(s) = C_T(1)$. The critical upper bound in Theorem 4.1, with $h = r$, also applies to $\mathcal{Q}_{s,1} \subseteq \mathcal{Q}_{s_*,1}$: $A_{N,T}(s) \leq C_1 e^{-c_1 T^{1/3}}$. Diagonal similarity transforms R to the symmetric killed simple-walk matrix on $1, \dots, N-1$, whose Euclidean norm is $\cos(\pi/N)$. The full-depth cover therefore gives

$$A_{N,T}(s) \leq e^{-\varepsilon T} e_1^{\mathsf{T}} R^T \mathbf{1} \leq \frac{\lambda}{\sqrt{\lambda^2 - 1}} e^{-\varepsilon T - \pi^2 T/(2N^2)}.$$

Taking the smaller upper bound gives $C_2 \exp[-c_2(T^{1/3} + \varepsilon T + T/N^2)]$. Since $A_{N,T}(s) \leq A_{N,1}(s) = e^{-\varepsilon} b \leq b < 1$, decreasing the exponent constant absorbs C_2 uniformly for $N \geq 3$, $T \geq 1$, and $s \in [s_*, 1]$.

For the lower bound, retain only paths within $1, \dots, m-1$, where $3 \leq m \leq N$, and define

$$\rho_m = \cos(\pi/m), \qquad \varphi_m(j) = \lambda^j \sin(j\pi/m), \qquad u(j) = \frac{\varphi_m(j)}{\max_{1\leq i<m} \varphi_m(i)}.$$

Extend u by zero outside this interval. Then $0 \leq u \leq 1$ and $Ru \geq \rho_m u$. Since $e^{-\varepsilon}\rho_m \leq 1$, induction in the capped recursion gives $C_T \geq e^{-\varepsilon T}\rho_m^T u$; capping at one preserves this lower bound at each step. Furthermore,

$$u(1) \geq \frac{2}{m}\lambda^{2-m}, \qquad \rho_m^T \geq e^{-\pi^2 T/m^2} \quad (m \geq 3).$$

Consequently

$$A_{N,T}(s) \geq \frac{2}{m}\lambda^{2-m} e^{-\varepsilon T - \pi^2 T/m^2}.$$

Choose $m = \min\{N, \max\{3, \lceil T^{1/3} \rceil\}\}$. For $T \leq N^3$ the last expression is at least $e^{-C(T^{1/3} + \varepsilon T)}$; for $T > N^3$ it is at least $e^{-C(N + \varepsilon T + T/N^2)}$. Since $N < T/N^2$ in the latter case, this proves (11), with $\varepsilon = (s - s_*)\log k$.

For the single-depth comparison, set $s = s_*$ and write S_N for the symmetric killed simple-walk matrix. Diagonal similarity gives

$$e_1^{\mathsf{T}} R^t \mathbf{1} = \sum_{j=1}^{N-1} \lambda^{1-j} S_N^t(1, j).$$

The reflection principle on the positive half-line gives, when parities match,

$$S^t_\infty(1,j) = 2^{-t}\left[\binom{t}{(t+j-1)/2} - \binom{t}{(t+j+1)/2}\right] \le C\frac{j}{(t+1)^{3/2}}.$$

For the inequality, factor the binomial difference as $2j/(t+j+1)$ times its first term and use the central-binomial bound. Since $S^t_N \le S^t_\infty$ and $\sum_{j\ge1} j\lambda^{1-j} < \infty$, the single-depth sum is at most $C(t+1)^{-3/2}$.

For even $t = 2\ell \le N-2$, the Catalan paths from 1 back to 1 stay below N. Their weight is $\mathrm{Cat}_\ell(ab)^\ell = \mathrm{Cat}_\ell/4^\ell \ge c(t+1)^{-3/2}$. For odd $t = 2\ell+1 \le N-2$, append one upward step to these paths; this multiplies the mass by b and gives the same order. Thus the single-depth sum is $\Theta_{k,r}((t+1)^{-3/2})$ for every $0 \le t \le N-2$.

It remains to compare covers of the same event $E_T = \{\tau > T\}$ at all depths, including depths exceeding T. Every live state has a live neighbor when $N \ge 3$, and hence has an arbitrarily long live continuation. For $t \le T$, the length-t cylinders meeting E_T are exactly the live words. Therefore $F_t(E_T) = e_1^\mathsf{T} R^t \mathbf{1}$. For $t > T$, all k^{t-T} extensions of each live length-T word belong to E_T, so

$$F_t(E_T) = k^{(1-s_*)(t-T)} e_1^\mathsf{T} R^T \mathbf{1} \ge F_T(E_T).$$

Taking the infimum over all depths proves (12). □

C.4 Sharp dependence on distance and source slack

The capped recursion retains a region where its value is exactly one. At the critical source exponent its boundary advances on the scale $t^{1/3}$; above the threshold there is an additional linear advance. The exact unit plateau will give a single admissible source that forces a prescribed delay with probability one.

In more detail, the boundary at distance m advances by order $\epsilon + m^{-2}$ per step, where $\epsilon = (s - s_*)\log k$. At criticality the speed is of order m^{-2}, so reaching distance j takes cubic time; a positive ϵ eventually gives speed of order ϵ and time j/ϵ. An exponentially tilted sine profile makes this comparison rigorous. Until the boundary reaches the initial distance, the capacity remains exactly one, already with $D = 1$, and tree max-flow supplies a source supported entirely on executions that survive the required time. For the slack dependence, a separate cover lower bound keeps capacity at least $1/D$ for the required horizon; the two constructions give $L = h + \log D + 1$ in Theorem 4.3.

Lemma C.3 (A slowly advancing plateau)**.** *Let $\lambda > 1$, $a = \lambda/2$, and $b = 1/(2\lambda)$. For $\epsilon \ge 0$ define, on the positive integers,*

$$C^\epsilon_0(j) = 1, \qquad C^\epsilon_t(0) = 0, \qquad C^\epsilon_{t+1}(j) = \min\{1, e^{-\epsilon}(aC^\epsilon_t(j-1) + bC^\epsilon_t(j+1))\}.$$

There are constants $\epsilon_0, c > 0$ and $j_0 \ge 1$, depending only on λ, such that, for $0 \le \epsilon \le \epsilon_0$,

$$C^\epsilon_t(j) = 1 \qquad \text{if} \qquad j \ge j_0, \qquad 0 \le t \le c\frac{j^3}{1+\epsilon j^2}.$$

In particular, the permitted horizon has order $\min\{j^3, j/\epsilon\}$, with $j/0 = \infty$.

The critical plateau order has an antecedent in Mallein [Mal19, Theorem 1.1]. Indeed, put $\beta = \log\lambda$ and take a branching random walk with bounded offspring, at least one child per particle, and expected child counts a, b at displacements $+\beta, -\beta$. Its boundary identities are $ae^{-\beta} + be^{\beta} = 1$

and $a\beta e^{-\beta} - b\beta e^{\beta} = 0$. The probability of a lineage staying below $j\beta$ through generation t is at most $C_t^0(j)$ by the union bound. Mallein's theorem makes this probability tend to one for $j = \lceil At^{1/3} \rceil \pm 1$ and sufficiently large A; since $a + b > 1$, the next capped step equals one. The crossover at time of order $\epsilon^{-3/2}$ is also analogous to the near-critical branching Brownian transition studied by Berestycki, Liu, Mallein, and Schweinsberg [BLMS24]. The proof below establishes the needed discrete capped-recursion bound directly; its application here is the construction of sources satisfying the prefix bound at every word.

Proof. Write $\beta = \log \lambda$. For each sufficiently large real m, let $\theta_m \in (0, \pi/m)$ be the unique solution of

$$m\theta_m + \arctan(\theta_m/\beta) = \pi.$$

Define a function on $[0, \infty)$ by

$$f_m(x) = \begin{cases} e^{\beta(x-m)} \dfrac{\sin(\theta_m x)}{\sin(\theta_m m)}, & 0 \le x < m, \\ 1, & x \ge m. \end{cases}$$

This function increases from zero to one: its logarithmic derivative below m is $\beta + \theta_m \cot(\theta_m x) > 0$, and that derivative vanishes at m. Write g_m for the sine expression without truncation and $\rho_m = \cos \theta_m$. Direct calculation gives

$$a g_m(x-1) + b g_m(x+1) = \rho_m g_m(x).$$

For $0 \le z \le 1$,

$$g_m(m+z) = e^{\beta z} \left(\cos(\theta_m z) - \frac{\beta}{\theta_m} \sin(\theta_m z) \right) \le 1;$$

its derivative in z is $-e^{\beta z}(\beta^2/\theta_m + \theta_m) \sin(\theta_m z) \le 0$. Thus replacing g_m by f_m cannot decrease a neighbor value used at an integer $j < m$. If $F_\epsilon f(j) = \min\{1, e^{-\epsilon}(af(j-1) + bf(j+1))\}$, then

$$F_\epsilon f_m(j) \ge e^{-\epsilon} \rho_m f_m(j) \qquad (1 \le j < m). \tag{25}$$

Choose fixed $z_0 \in (0, 1)$ and $\epsilon_0 > 0$ sufficiently small that

$$e^{-\epsilon_0} \left[a e^{-\beta(z_0+1)} (1 + \beta(z_0+1)) + b \right] > 1.$$

Such a choice is possible because the bracket at $z_0 = 0$ equals $(1 + \beta + e^{-\beta})/2 > 1$. For every fixed $z \ge 0$, $f_m(m - z) \longrightarrow e^{-\beta z}(1 + \beta z)$. Consequently, for all sufficiently large m and uniformly in $0 \le \epsilon \le \epsilon_0$,

$$F_\epsilon f_m(j) = 1 \qquad (j \ge m - z_0). \tag{26}$$

Indeed, monotonicity gives $f_m(j-1) \ge f_m(m - z_0 - 1)$, while $f_m(j+1) = 1$.

Away from this saturated band, increasing m decreases f_m at a uniform logarithmic rate. Differentiating the equation for θ_m gives

$$-\frac{\theta_m'}{\theta_m} = \frac{1}{m + \beta/(\beta^2 + \theta_m^2)}.$$

For fixed $0 < x < m$ it follows that

$$-\partial_m \log f_m(x) = \frac{\theta_m x \cot(\theta_m x) + \beta m}{m + \beta/(\beta^2 + \theta_m^2)}. \tag{27}$$

Since $y \cot y$ decreases on $(0, \pi)$, the right side for $x \le m - z_0$ is bounded below by its value at $x = m - z_0$. That value converges to $\beta^2 z_0/(1 + \beta z_0) > 0$. Hence there are $m_0 \ge 1$ and $\eta > 0$ such that $-\partial_m \log f_m(x) \ge \eta$ whenever $m \ge m_0$ and $0 < x \le m - z_0$. Enlarge m_0 so that all the preceding comparisons hold and $\rho_m > 0$ for $m \ge m_0$.

For fixed $\epsilon \in [0, \epsilon_0]$, set

$$\Delta_m = \frac{\epsilon - \log \rho_m}{\eta}.$$

For $j < m - z_0$, integrating (27) over $[m, m + \Delta_m]$ gives

$$f_{m+\Delta_m}(j) \le e^{-\epsilon} \rho_m f_m(j) \le F_\epsilon f_m(j).$$

For the remaining j, use (26). Thus $F_\epsilon f_m \ge f_{m+\Delta_m}$ pointwise. Since $\theta_m < \pi/m$, there is a constant A such that, uniformly in $m \ge m_0$ and $\epsilon \in [0, \epsilon_0]$,

$$0 < \Delta_m \le A(\epsilon + m^{-2}).$$

Starting from m_0, define $m_{t+1} = m_t + \Delta_{m_t}$. Monotonicity of F_ϵ and $C_0^\epsilon \ge f_{m_0}$ give $C_t^\epsilon \ge f_{m_t}$ by induction.

We now bound this discrete sequence. Put $Q = A(\epsilon_0/m_0 + m_0^{-3})$ and $K = A(3 + 3Q + Q^2)$. Because $\Delta_m/m \le Q$,

$$(m + \Delta_m)^3 - m^3 \le (3 + 3Q + Q^2) m^2 \Delta_m \le K(1 + \epsilon m^2).$$

For any $j \ge 2m_0$, suppose there is a first index $u \le t$ with $m_u > j$. The preceding values are at most j, so summing the last inequality yields

$$j^3 < m_u^3 \le m_0^3 + Ku(1 + \epsilon j^2) \le m_0^3 + Kt(1 + \epsilon j^2).$$

This is impossible if $t \le cj^3/(1 + \epsilon j^2)$, where $c \le 1/(2K)$, because $m_0^3 \le j^3/8$. Therefore $m_t \le j$ throughout that horizon and $C_t^\epsilon(j) \ge f_{m_t}(j) = 1$. Taking $j_0 = \lceil 2m_0 \rceil$ proves the lemma. □

Proof of Theorem 4.3. Write $\delta = s - s_*$, $\epsilon = \delta \log k$, $\ell = \log D$, and $j = h - r + 1 \ge 1$. The source-label sequence determines the distance process of Φ_n independently of the clause selector. For an integer horizon $T \ge 0$, choose $N = j + T + 2$ and $n = N + k - 2$. No path starting at j reaches the upper boundary by time T. Since $R_s = e^{-\epsilon} R_{s_*}$, the finite-interval capacity at (j, T) therefore equals $C_T^\epsilon(j)$. The finite-tree capacity identity, with root capacity one and all nonroot capacities multiplied by D, gives

$$\max_{\nu \in \mathcal{Q}_{s,D}} \Pr_\nu\{\tau > T\} = \min\{1, DC_T^\epsilon(j)\}. \tag{28}$$

The maximum is attained. A maximizing finite-depth law extends to an infinite admissible source by appending independent uniform symbols: for $t \ge T$, its prefix bound is $Dk^{-sT}k^{-(t-T)} \le Dk^{-st}$ because $s \le 1$.

First suppose $0 \le \epsilon \le \epsilon_0$. Lemma C.3 gives $C_T^\epsilon(j) = 1$ whenever $j \ge j_0$ and

$$T \le c \frac{j^3}{1 + \epsilon j^2}.$$

This supplies a probability-one delay of order $\min\{j^3, j/\epsilon\}$ for large j. To treat slack, monotonicity of the recurrence shows that $C_t^\epsilon(j)$ is increasing in j. The elementary inequality $\min\{1, \alpha x\} \ge \alpha \min\{1, x\}$ for $0 \le \alpha \le 1$ also gives, by induction,

$$C_T^\epsilon(1) \ge e^{-\epsilon T} C_T^0(1).$$

The fixed-strip eigenvector lower bound used in Theorem 4.2(ii), with width $\max\{3, \lceil T^{1/3} \rceil\}$, yields

$$C_T^{\epsilon}(1) \geq \exp[-\epsilon T - B(T+1)^{1/3}]$$

for a constant B depending only on k, r. For all sufficiently large ℓ, choose

$$T = \left\lfloor c' \min\{\ell^3, \ell/\epsilon\} \right\rfloor,$$

again interpreting division by zero as infinity. Choose $c' > 0$ so that $c' \leq 1/4$ and $B(c')^{1/3} \leq 1/4$. Then

$$\epsilon T + B(T+1)^{1/3} \leq c'\ell + B(c')^{1/3}\ell + B \leq \ell$$

for sufficiently large ℓ. Thus $DC_T^{\epsilon}(j) \geq 1$, and (28) supplies another probability-one delay, now of order $\min\{\ell^3, \ell/\epsilon\}$.

These two horizons combine uniformly. For $f_\epsilon(x) = \min\{x^3, x/\epsilon\}$, monotonicity and $f_\epsilon(ux) \leq u^3 f_\epsilon(x)$ for $u \geq 1$ show that the larger horizon, plus one to cover rounding and bounded parameters, is at least $c_1 f_\epsilon(h + \ell + 1)$. Here $h/r \leq j \leq h$, and $h \geq r$, so the change from j to h costs only a constant depending on r. Because $\epsilon = (s - s_*) \log k$, $f_\epsilon(L) = \Theta_k(\Psi_s(L))$ for $L = h + \ell + 1$. Reducing a fixed constant $c_2 > 0$ therefore gives an admissible source with

$$\Pr_{\nu}\{\tau > \lfloor c_2 \Psi_s(L) \rfloor\} = 1.$$

For bounded parameters the displayed horizon is zero and the initial assignment is live; this also accounts for all floor terms.

It remains to treat $\epsilon \geq \epsilon_0$. The target satisfies $\Psi_s(L) \leq (\log k/\epsilon_0)L$. Distance alone gives survival through $j - 1$ steps under any source, provided the upper boundary is farther than that horizon. For slack, let

$$T_D = \left\lfloor \frac{\ell}{s \log k} \right\rfloor.$$

Use a fixed upward label for the first T_D symbols and independent uniform symbols thereafter. Up to depth T_D, every nonzero cylinder has probability one and satisfies $1 \leq Dk^{-st}$; after that depth, the same extension calculation as above proves all prefix bounds. With $N > j + T_D$ this source forces $\tau > T_D$. The larger of $j - 1$ and T_D, plus one, is at least a constant times $h + \ell + 1$. Reducing c_2 again proves the same probability-one delay throughout the remaining range of s.

Finally take $T = \lfloor c_2 \Psi_s(L) \rfloor$ and $N = j + T + 2$; any source constructed for a longer horizon can be restricted to depth T and extended uniformly. The number of variables is $n = j + T + k = O_{k,r}(\Psi_s(L))$, since $\Psi_s(L) \geq L \geq h$ for $s \in [s_*, 1]$. Moreover $\tau > T$ implies $\tau \geq T + 1 > c_2 \Psi_s(L)$ almost surely, and hence gives the same lower bound on its expectation. All claims hold simultaneously for every selector under the same source because the label-to-distance map is selector independent. □

C.5 An observable restart rule for the promise-SAT walk

Proof of Theorem 4.4. Use the unknown witness distance $d(x) = \mathrm{dist}(x, x^*)$ only in the analysis. Suppose a proposed flip reaches an unsatisfying assignment y outside the ball of radius $2H$ around x_0. The triangle inequality gives

$$d(y) \geq \mathrm{dist}(y, x_0) - \mathrm{dist}(x_0, x^*) \geq 2H + 1 - h \geq h + 1.$$

Resetting to x_0 therefore strictly decreases this potential. Consequently, after the flip and any reset, each of the at least r favorable symbols still either terminates the walk or decreases d by at least

one; every other symbol increases d by at most one. Every live assignment kept by the algorithm lies within distance $2H$ of x_0, so its potential is at most $L = h + 2H$.

Let $a = \lambda/2$ and $b = 1/(2\lambda)$ as in Lemma C.1. On $\{1, \dots, L\}$ define the reflected comparison matrix

$$(Rf)(d) = af(d-1) + bf(d+1) \quad (d < L), \qquad (Rf)(L) = af(L-1) + bf(L),$$

where $f(0) = 0$. Its powers preserve nonnegative increasing vectors. Exactly as in the barrier lemma, induction on the remaining horizon therefore bounds the weight of surviving length-T execution words by $e_h^{\mathsf{T}} R^T \mathbf{1}$. This comparison is valid after every history, including histories containing resets; the clause selector may retain the entire history. The spectral estimate in that lemma gives

$$e_h^{\mathsf{T}} R^T \mathbf{1} \le \frac{\lambda^h}{\sqrt{\lambda^2 - 1}} \exp\left(-\frac{\pi^2 T}{2(2L+1)^2}\right).$$

Apply the source constraint to whole length-T execution words and sum. Since $L = h + 2H \le 3H$, the resulting bound is at most $CD\lambda^h e^{-cT/H^2}$, proving (13) for $s = s_*$; the inclusion $\mathcal{Q}_{s,D} \subseteq \mathcal{Q}_{s_*,D}$ gives the claim for $s \ge s_*$.

Since $L \le 3H$, summing the capped exponential tail yields $O_{k,r}(H^2(h + \log D + 1))$. More explicitly, for $A > 0$ and $q > 0$, $\sum_{T \ge 0} \min\{1, Ae^{-qT}\}$ is at most $1 + q^{-1}(1 + \max\{0, \log A\})$ by comparison with the integral of the decreasing summand. Here $q^{-1} = O(H^2)$ and $\log A = h \log \lambda + \log D - \frac{1}{2}\log(\lambda^2 - 1)$.

For the implementation claim, keep a list of coordinates on which the current assignment differs from x_0, together with an inverse index for constant-time insertion and deletion. A flip updates this list in constant time. At a reset, restore and remove every listed coordinate. Each restored coordinate was flipped at least once in that attempt, so the reset uses no more variable updates than the attempt used source-driven flips. Summing over disjoint attempts proves the factor-two bound. The list has size at most $2H + 1$ before a reset. □

C.6 A radius-free portfolio using an archived tape

Theorem C.4 (Unknown radius with an archived tape)**.** *There is a deterministic algorithm that requires neither a distance bound nor the source slack D and has the following guarantee. Suppose a strongly r-satisfiable k-CNF on n variables has a strongly satisfying assignment at distance $h \ge 1$ from the initial assignment. Put $\ell_h = 1 + \lceil \log_2 h \rceil$ and $L = h + \log D + 1$. For every $D \ge 1$, $s \in [s_*, 1]$, and $\nu \in \mathcal{Q}_{s,D}$, the algorithm's total number σ of variable updates satisfies, for every integer $T \ge 0$,*

$$\mathbb{P}_\nu\{\sigma > T\} \le \min\left\{1, CD\lambda^h \exp\left(-\frac{cT}{h^2 \ell_h^2}\right)\right\},$$

and

$$\mathbb{E}_\nu \sigma \le C \min\left\{\Psi_s(L), h^2 \ell_h^2 L\right\}.$$

Here $c, C > 0$ depend only on k, r. The update count includes every flip in every simulation and every coordinate restored during a reset. The algorithm keeps at most $2 + \lceil \log_2 n \rceil$ simulated assignments and at most σ archived source symbols. Each simulation may use any deterministic clause selector based only on its own local history. Work to find a violated clause is separate.

The algorithm. Stop immediately if the initial assignment x_0 satisfies the formula. Let $J = \lceil \log_2 n \rceil$. Keep one base simulation of the original walk and auxiliary simulations A_j, $0 \le j \le J$, of the Hamming-ball restart rule with radius parameter $H_j = 2^j$. All simulations start from x_0 and use the same source tape $\omega_1, \omega_2, \ldots$; an auxiliary simulation can be initialized when first scheduled. Partition the nonnegative indices into groups

$$G_b = \{2^b - 1, \ldots, 2^{b+1} - 2\}, \qquad b \ge 0.$$

In round m, first select the base walk's clause using its past history; then read and archive ω_m and perform the indexed flip. Then choose the group G_b with $b = v_2(m)$, the largest integer with 2^b dividing m, and select its next index in cyclic order, beginning at its smallest index. If the selected index is at most J, advance that auxiliary simulation once, including its prescribed reset if needed. Otherwise leave the auxiliary slot unused. On its qth activation, an auxiliary simulation uses the archived symbol ω_q. Halt as soon as any simulation satisfies the formula.

Every auxiliary activation has $q \le m$, so the required symbol is already available. Before its qth flip, its clause selector may use only that simulation's own states and $\omega_1, \ldots, \omega_{q-1}$; it must not inspect ω_q or any later archived symbol. For example, every simulation can always choose the first violated clause in a fixed ordering. Full assignment arrays require $O(n \log(n+2))$ walk-state storage in addition to the archive and any selector workspace. Simulations may instead be initialized lazily relative to x_0.

Proof of Theorem C.4. Define counterfactual complete simulations on the full tape, even if the combined algorithm halts earlier. Let τ_{base} be the base walk's local stopping time and let τ_j be that of A_j, counting its source-driven flips. Each simulation reads the original tape prefix in its original order. Its tape law is therefore exactly ν, so Theorems 4.1 and 4.4 apply to the respective simulations. Independence between them is not needed.

Let $N_j(M)$ be the number of activations assigned to A_j in the first M rounds. If $j \in G_b$, put $z = j - (2^b - 1)$ and $q_j = 2^{2b+1}$. The activation rounds of this index are exactly

$$m \equiv 2^b(1 + 2z) \pmod{q_j}, \qquad q_j \le 2(j+1)^2.$$

Since the residue lies strictly between zero and q_j,

$$N_j(M) \ge \lfloor M/q_j \rfloor \ge M/q_j - 1. \tag{29}$$

By round M the algorithm has made at most M base flips and M auxiliary source-driven flips. In each auxiliary simulation, the number of restored coordinates over all resets is at most the number of its source-driven flips: each restored coordinate was flipped during that attempt, and attempts are disjoint. The total number of variable updates by round M, or by earlier termination, is thus at most $3M$. In particular, $\sigma \le 3\tau_{\text{base}}$ pathwise. Theorem 4.1 therefore gives $\mathbb{E}_\nu \sigma \le C\Psi_s(h + \log D + 1)$.

Choose $j_* = \lceil \log_2 h \rceil$. Then $j_* \le J$ and $h \le H_{j_*} < 2h$. If $\sigma > T$, the combined algorithm has not terminated within the first $M = \lfloor T/3 \rfloor$ rounds, since termination in those rounds would use at most $3M \le T$ updates. Consequently

$$\{\sigma > T\} \subseteq \{\tau_{j_*} > N_{j_*}(\lfloor T/3 \rfloor)\}.$$

Write c_0, C_0 for the constants in Theorem 4.4. From (29), $N_{j_*}(\lfloor T/3 \rfloor) \ge T/(3q_{j_*}) - 2$. Applying the known-radius bound to this one simulation gives

$$\begin{aligned}
\mathbb{P}_\nu\{\sigma > T\} &\le C_0 e^{2c_0} D\lambda^h \exp\left(-\frac{c_0 T}{3q_{j_*} H_{j_*}^2}\right) \\
&\le C_0 e^{2c_0} D\lambda^h \exp\left(-\frac{c_0 T}{24h^2(j_*+1)^2}\right).
\end{aligned}$$

This proves the claimed tail after capping by one. Summing that capped exponential gives the second expectation bound. Taking the smaller of the two expectation estimates proves the result. Finally, each round reads one new symbol and performs at least one base update before any auxiliary work. Hence at termination the archive contains at most σ symbols. □

D Proofs for restart policies

D.1 Necessity in Theorem 6.1

The proof in the main text establishes (i)⇒(ii), and (ii)⇒(iii) is immediate. Suppose now that $\alpha_s(x) = 1$. Exact prefix capacity gives a probability law $\mu \in \mathcal{Q}_{s,1}$ supported on $\mathcal{N}_x$. Fix an arbitrary deterministic restart policy. At each restart, begin an independent stream with law μ. This defines an adversarial source for the entire execution, not an assumption about the algorithm's access to independent randomness.

For any observed word w, the policy determines a unique segmentation into completed attempt words $b_1, \dots, b_j$ and an unfinished attempt word v. The resulting source therefore satisfies

$$\nu[w] = \mu[b_1] \cdots \mu[b_j]\mu[v] \le P[b_1]^s \cdots P[b_j]^s P[v]^s = P[w]^s.$$

The formula is consistent on the prefix tree, including at restart boundaries. Every attempt follows an original nonterminating execution until the policy abandons it. Every prefix on which an attempt terminates has probability zero. The countable union of these prefixes still has probability zero, proving the converse and the stronger final assertion.

For completeness, the time bound in the sufficiency proof uses the horizon T, rather than the longest failed-attempt word: a successful path can avoid $\mathcal{B}$ and run longer than every word in that cover. If the longest cover word has length L, the slightly sharper bound $\Pr_\nu\{\tau_{\text{restart}} > T+kL\} \le Dc^k$ also holds.

D.2 Proof of Theorem 6.2

The greatest feasible vector. The finite-horizon controlled recursion of Proposition B.2 is

$$V_0 = \mathbf{1}, \qquad V_{t+1}(y) = \min\left\{1, \min_{f \in F_y} \sum_{z \in X} R_f(y,z) V_t(z)\right\}.$$

The sequence decreases, and its limit V is a fixed point because the state and action sets are finite. Every feasible vector v in (15) satisfies $v \le V_t$ by induction, hence $v \le V$. Conversely, the fixed point V is feasible. Thus V is the componentwise greatest feasible vector and the unique maximizer of the sum objective.

For every fixed history-dependent selector Λ, finite-horizon optimality gives $\mathcal{C}_s(\{\tau_\Lambda > t\}) \ge V_t(x)$. Continuity in Theorem 2.1 therefore gives

$$\mathcal{C}_s(\{\tau_\Lambda = \infty\}) \ge V(x). \tag{30}$$

This argument fixes Λ before taking the decreasing-event limit; it does not interchange an infimum over selectors with that limit.

Selecting a stationary rule. Set $U = \{y : V(y) < 1\}$ and $S = X \setminus U$. For $y \in U$, let

$$I_y = \left\{ f \in F_y : \sum_{z \in X} R_f(y,z) V(z) = V(y) \right\}.$$

Each I_y is nonempty. On U, solve the second linear program

$$\max \left\{ \sum_{y \in U} \psi(y) : \psi \geq 0, \quad \psi(y) \leq 1 + \sum_{z \in U} R_f(y,z)\psi(z) \text{ for every } y \in U,\ f \in I_y \right\}. \tag{31}$$

If U is empty, this construction is vacuous and the selector may be chosen arbitrarily.

The second program is feasible and bounded. Indeed, a nonzero recession direction would be a vector $w \geq 0$ satisfying

$$w(y) \leq \sum_{z \in U} R_f(y,z) w(z) \qquad (y \in U,\ f \in I_y).$$

Extend w by zero on S. For sufficiently small $\epsilon > 0$, $V + \epsilon w$ remains below $\mathbf{1}$ on U and preserves every constraint of (15): tight constraints on U use the displayed inequalities, nontight constraints have positive slack, and constraints on S can only be relaxed. This contradicts the componentwise maximality of V. Thus there is no nonzero recession direction, and the feasible polyhedron is bounded.

At an optimizer ψ, every state $y \in U$ has a binding action in (31). Otherwise increasing $\psi(y)$ slightly preserves its finitely many strict inequalities and relaxes all inequalities belonging to other states, contradicting optimality. Choose one binding action $\mu(y) \in I_y$, and choose μ arbitrarily on S. Define

$$A(y,z) = R_{\mu(y)}(y,z) \quad (y,z \in U), \qquad b(y) = \sum_{z \in S} R_{\mu(y)}(y,z).$$

The binding constraints and membership in I_y give

$$\psi = \mathbf{1} + A\psi, \qquad V_U = AV_U + b. \tag{32}$$

In particular $\psi \geq \mathbf{1}$ and $A\psi < \psi$, so $\rho(A) < 1$. Consequently,

$$\psi = (I - A)^{-1}\mathbf{1}, \qquad V_U = (I - A)^{-1} b.$$

This second program is needed only to resolve ties properly. For example, at a state of value zero, both an update that always terminates and an update that always returns to the same state are tight at V; the latter is not a valid choice for synthesis.

Exact optimality without restarts. Let W be the nontermination capacity vector of the fixed stationary selector μ. Equivalently, W is the greatest feasible vector for the fixed-matrix constraints $0 \leq w \leq \mathbf{1}$, $w \leq R_\mu w$, by the same decreasing-recursion argument. Since V is feasible for these constraints, $W \geq V$ and hence $W_S = \mathbf{1}$. On U,

$$W_U \leq AW_U + b, \qquad W_U \leq (I - A)^{-1} b = V_U.$$

Thus $W = V$. Combining this with (30) and the exact capacity identity proves that μ attains the stated minimum, simultaneously for every initial state and every $D \geq 1$.

Restart tail bound. Suppose $x_0 \in U$. Under μ and immediate restart on entry to S, the powered live matrix on U is

$$B = A + be_{x_0}^{\mathsf{T}}.$$

Set

$$g = \psi + \frac{\psi(x_0)}{1 - V(x_0)} V_U, \qquad H = \max_{y \in U} g(y).$$

Equation (32) gives $Bg = g - \mathbf{1}$. Therefore $g \geq \mathbf{1}$, $\rho(B) < 1$, and $g = (I - B)^{-1}\mathbf{1}$. If $H > 1$, then $Bg \leq (1 - H^{-1})g$; if $H = 1$, then $B\mathbf{1} = 0$ and termination occurs after one source-consuming step.

Summing powered probabilities over complete surviving words gives

$$\mathbb{P}_\nu\{\tau_{\text{restart}} > t\} \leq \min\{1, De_{x_0}^{\mathsf{T}} B^t \mathbf{1}\}, \tag{33}$$

$$\mathbb{E}_\nu \tau_{\text{restart}} \leq Dg(x_0) = D\,\frac{\psi(x_0)}{1 - V(x_0)} = D\,\frac{e_{x_0}^{\mathsf{T}}(I - A)^{-1}\mathbf{1}}{1 - V(x_0)}. \tag{34}$$

For $H > 1$, the first bound is at most $Dg(x_0)(1 - H^{-1})^t$. The mean bound follows by summing the matrix bound over $t \geq 0$. These estimates invoke the source constraint only on prefixes of the whole execution tape; restarting does not restore a conditional source guarantee.

Impossibility for arbitrary controlled restart policies. Suppose $V(x_0) = 1$, and fix any deterministic nonanticipating policy that chooses updates and may restart to x_0 between successive steps. We construct one source in $\mathcal{Q}_{s,1}$ for this entire policy. Write $V = 0$ at terminated states. At a positive-probability history whose current live state is y and chosen update is f, set

$$L(y, f) = \sum_a \pi(a)^s V(\mathcal{T}_f(y, a)) \geq V(y) > 0, \qquad q(a \mid \text{history}) = \frac{\pi(a)^s V(\mathcal{T}_f(y, a))}{L(y, f)}.$$

These probabilities sum to one. They assign zero probability to termination and to states of value zero. Within any attempt, their product along a word b ending at z is at most $P[b]^s V(z)/V(x_0) = P[b]^s V(z)$, by telescoping the successive values and using $L(y, f) \geq V(y)$.

For any prefix w of the execution tape, the policy determines its unique segmentation into completed attempts and a current partial attempt. Multiplying the preceding bounds over these segments gives $\nu[w] \leq P[w]^s$, since each endpoint value is at most one. Thus the constructed law lies in $\mathcal{Q}_{s,1}$ and has nontermination probability one. Its definition may depend on the fixed policy, as permitted by the minimax quantifiers. This proves necessity even for policies that retain their entire histories across restarts.

Exact computation. If the powered coefficients $R_f(y, z)$ are rational, the first linear program, its unique optimizer, the tight action sets, and the second linear program all have polynomial-size rational descriptions. Exact linear programming and rational linear algebra therefore compute μ, S, and g in polynomial bit complexity in the explicit state–action input. The resulting execution-time constants need not be polynomially bounded. Approximate representations of arbitrary real values $\pi(a)^s$ do not by themselves provide the exact rational input required by this assertion.

D.3 Adaptive versus predetermined restarts

Proposition D.1 (Adaptive versus predetermined restarts)**.** *A three-live-state procedure has reference mean running time* $8/3$ *under a uniform four-symbol source and the following properties. For every predetermined sequence of positive integer cutoffs, allowing a final infinite cutoff, some source in* $\mathcal{Q}_{1/2,1}$ *gives nontermination probability at least* $1/2$*. In contrast, one stationary restart rule satisfies, for every* $D \geq 1$ *and integer* $T \geq 2$*,*

$$\max_{\nu \in \mathcal{Q}_{1/2,D}} \Pr_{\nu}\{\tau_{\text{adaptive}} > T\} = \min\{1, D\,T2^{1-T}\}.$$

Proof. Use source alphabet $\{0,1,2,3\}$, initial state x, and the transition table below, where † denotes termination.

	0	1	2	3
x	†	y	z	z
y	†	y	y	y
z	z	†	†	†

Under the uniform product law,

$$\mathbb{E}_P \tau = 1 + \tfrac{1}{4} \cdot 4 + \tfrac{1}{2} \cdot \tfrac{4}{3} = \tfrac{8}{3}.$$

At exponent $1/2$, each symbol has weight $1/2$. The original powered live matrix, in state order (x, y, z), is

$$R = \begin{pmatrix} 0 & 1/2 & 1 \\ 0 & 3/2 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}.$$

For every $T \geq 1$, the failed length-T words F_T have total weight

$$Z_T^{\text{live}} := \sum_{w \in F_T} P[w]^{1/2} = \tfrac{1}{2}(3/2)^{T-1} + (1/2)^{T-1} \geq 1.$$

For $T = 1, 2$, the values are $3/2, 5/4$; for $T \geq 3$, the first term alone exceeds one. The original nontermination capacities are $(1/2, 1, 0)$.

For a finite cutoff T, put the probability law $q_T(w) = P[w]^{1/2}/Z_T^{\text{live}}$ on F_T. At a nonempty partial word u of length $j \leq T$, the ratio $q_T[u]/P[u]^{1/2}$ is zero or one of

$$\frac{(3/2)^{T-j}}{Z_T^{\text{live}}} \leq 2 \quad \text{on the } y \text{ branch}, \qquad \frac{(1/2)^{T-j}}{Z_T^{\text{live}}} \leq 1 \quad \text{on the } z \text{ branch}.$$

At a complete block, this ratio is $1/Z_T^{\text{live}} \leq 1$. For any predetermined finite-cutoff sequence, concatenate independent blocks with these laws. Each complete block contributes a ratio at most one, and the final partial block contributes at most two. The resulting law $\mu \in \mathcal{Q}_{1/2,2}$ is supported on nontermination of the scheduled procedure. If the schedule has a final infinite cutoff, choose its first symbol to be 1, then choose independent uniform symbols from $\{1,2,3\}$ forever. The partial-prefix ratios in this final attempt are also at most two. Hence $\mu/2$ is a subprobability measure satisfying the unit prefix caps. Prefix-flow completion extends it to a probability law in $\mathcal{Q}_{1/2,1}$ while retaining nontermination mass at least $1/2$. The adversarial source may depend on the predetermined schedule.

The adaptive rule restarts immediately on entry to y. Its powered live matrix on (x, z) is $B = \left(\begin{smallmatrix} 1/2 & 1 \\ 0 & 1/2 \end{smallmatrix}\right)$. The exact capped recursion gives $v_T(z) = 2^{-T}$ and $v_0(x) = v_1(x) = v_2(x) = 1$. For $T \geq 3$,

$$v_T(x) = \tfrac{1}{2} v_{T-1}(x) + 2^{-(T-1)} = T2^{1-T}.$$

The formula also holds at $T = 2$. Exact prefix capacity now gives $\max_{\nu \in \mathcal{Q}_{1/2,D}} \Pr_{\nu}\{\tau_{\text{adaptive}} > T\} = \min\{1, D\,T2^{1-T}\}$. □

E Proofs for scheduling

E.1 Optimal priority for heterogeneous retry tasks

Proof of Theorem 3.1. For a set S of unfinished tasks, the controlled recursion gives

$$V_{t+1}(S) = \min\left\{1, \min_{i\in S}((W-w_i)V_t(S) + w_i V_t(S\setminus\{i\}))\right\}, \tag{35}$$

with $V_t(\varnothing) = 0$ and $V_0(S) = 1$ for nonempty S. Symbols with the same success status lead to the same state, so their weights add even when a selector remembers the individual labels. Proposition B.2 gives the exact optimum $\min\{1, DV_T(S)\}$.

We first record three monotonicities. Values decrease with t, because the recursion operator is monotone and $V_1 \le V_0$. They increase when tasks are added. Indeed, suppose $R \subseteq S$ and argue by induction on t. An action $i \in R$ has at least its corresponding cost on R. An action $i \in S \setminus R$ has cost at least $WV_t(R) \ge V_t(R) \ge V_{t+1}(R)$, since $W \ge 1$. Taking the minimum and truncating at one preserve the comparison.

For fixed W, increasing a task's weight cannot increase the value. To see this, compare recursions in which its weight is a and $b \ge a$. By induction, the present-state values satisfy $X_a \ge X_b$, and the common value after completing that task is $Y \le X_b$. The corresponding action costs differ by

$$(W-a)X_a + aY - ((W-b)X_b + bY) = (W-a)(X_a - X_b) + (b-a)(X_b - Y) \ge 0.$$

Other action costs are ordered by induction. These arguments apply to the abstract recursion for all $0 < w_i \le W$ and $W \ge 1$, without requiring a modification of the reference source.

Subtracting the costs of two possible first tasks cancels the common W term. What remains is a comparison of the values obtained by removing either task. This motivates the following stronger invariant, which we prove by induction on t: for distinct $i, j \in S$ with $a = w_i < b = w_j$,

$$(b-a)V_t(S) \le bV_t(S\setminus\{j\}) - aV_t(S\setminus\{i\}). \tag{36}$$

It holds at $t = 0$. The inequality says exactly that the action for i has no larger cost than the action for j in (35). Equal-weight tasks are interchangeable. Thus, if the invariant holds at time t, a minimum-weight task is an optimal first action for horizon $t+1$.

We shall use the following elementary truncation fact. If $0 \le z \le y \le x$ and $(b-a)x \le by - az$, then

$$(b-a)\min\{1,x\} \le b\min\{1,y\} - a\min\{1,z\}. \tag{37}$$

If $x \le 1$, nothing changes. If $y \ge 1$, the right side is at least $b-a$. Otherwise $x > 1 > y$, and the unchanged right side is at least $(b-a)x \ge b-a$.

Fix i, j as above, and choose a minimum-weight task $k \in S$. First suppose $k \notin \{i, j\}$ and put $c = w_k$. The optimal first action in S, $S\setminus\{i\}$, and $S\setminus\{j\}$ can be k. Before truncation, each value is the same nonnegative linear combination, with coefficients $W-c, c$, of the corresponding values with and without k at time t. Combine (36) for S and $S\setminus\{k\}$ to obtain the desired inequality before truncation. The three quantities have order $x \ge y \ge z$: adding a task increases the value, and replacing a weight a by b decreases it. Equation (37) proves the invariant after truncation.

The remaining case is $k = i$, since $a < b$ excludes $k = j$. Write

$$X = V_t(S), \quad Y = V_t(S\setminus\{j\}), \quad Z = V_t(S\setminus\{i\}), \quad H = V_t(S\setminus\{i,j\}).$$

The monotonicities give $X \ge Y \ge Z \ge H$. Set

$$x = (W-a)X + aZ, \qquad y = (W-a)Y + aH, \qquad z = (W-b)Z + bH.$$

The first two quantities are the optimal costs before truncation for S and $S \setminus \{j\}$. The third is a permissible cost for $S \setminus \{i\}$, obtained by choosing j. Hence its truncated value is an upper bound on $V_{t+1}(S \setminus \{i\})$. We have $x \geq y$ and

$$y - z = (W - a)(Y - Z) + (b - a)(Z - H) \geq 0.$$

Furthermore,

$$by - az - (b - a)x = (W - a)\big(bY - aZ - (b - a)X\big) \geq 0.$$

Apply (37), then replace its last truncated quantity by the no-larger actual value. This proves (36) at $t + 1$ and completes the induction. Thus increasing-weight priority is optimal at every horizon. Following that order leaves only suffixes of the sorted list and gives the stated $O(nT)$ recursion. Monotonicity of $x \mapsto \min\{1, Dx\}$ proves simultaneous optimality for every $D \geq 1$.

Finally, fix $\theta > 0$, let $W_\theta = \sum_a \pi(a)^\theta$, and consider the tilted law $\pi_\theta(a) = \pi(a)^\theta / W_\theta$. Under its independent stream, preassign independent streams to the tasks. The number of attempts for task i is geometric with success probability $\sum_{a \in B_i} \pi_\theta(a)$. The workloads are independent and finite almost surely. Every permitted selector uses exactly their sum: before all tasks complete it must attempt some unfinished task, and none can receive more than its preassigned workload. The next unrevealed symbol of any selected stream has law π_θ, even when the selection uses all previous observations. The stopping-time distribution is therefore selector independent. Multiplication by W_θ^T proves equality of the prefix power sums $\widetilde{Z}_T(\theta)$, and the identity $Z_T(\theta) = W_\theta \widetilde{Z}_{T-1}(\theta)$ proves it for $Z_T(\theta)$. □

E.2 Equal stopping laws and different robustness

Proof of Theorem 3.2. Put

$$u = \sqrt{\pi(0)} = \frac{\sqrt{15}}{4}, \qquad v = \sqrt{\pi(1)} = \frac{1}{4}.$$

We first construct ν_n. Through time $8n$, generate actions with probabilities $3/4$ for 0 and $1/4$ for 1 until the nth zero appears; after that zero, output only zeros through time $8n$. After time $8n$, generate independent actions according to P. This specifies one law on tapes, without reference to the selector.

A prefix of length at most $8n$ preceding the nth zero, with k zeros, satisfies

$$\frac{\nu_n[w]}{P[w]^{1/2}} = \left(\frac{3}{\sqrt{15}}\right)^k \leq 1.$$

Suppose the nth zero appears at time $L \leq 8n$. For a positive-mass prefix of length $t \in [L, 8n]$, the corresponding ratio is

$$\left(\frac{3}{\sqrt{15}}\right)^n u^{-(t-L)} \leq \left(\frac{3/4}{u^8}\right)^n < 1,$$

since $L \geq n$ and $u^8 = (15/16)^4 > 3/4$. Zero-mass prefixes satisfy the required bound automatically. After time $8n$, each new reference action multiplies the ratio by $\pi(a)^{1/2} \leq 1$. Thus every finite prefix satisfies the defining inequality for $\mathcal{Q}$. Under Λ_A, either fewer than n zeros have appeared by time $8n$, so an A task remains unfinished, or the B task is attempted only after the nth zero and subsequently sees zeros alone. In both cases $\tau > 8n$, proving (6).

For (7), consider an arbitrary $\nu \in \mathcal{Q}$ and use Λ_B. If no 1 appears during the first $4n$ actions, the tape begins with 0^{4n}, an event of ν-probability at most

$$u^{4n} = \left(\frac{225}{256}\right)^n.$$

Otherwise let $j \le 4n$ be the first position containing 1. This completes B, and survival after $8n$ repairs requires fewer than n zeros among the remaining $\ell = 8n - j \ge 4n$ actions. Set $r = v/(3u) = 1/(3\sqrt{15})$. Their total power weight is at most

$$\begin{aligned}\sum_{k=0}^{n-1} \binom{\ell}{k} u^k v^{\ell-k} &\le r^{-(n-1)}(v + ur)^\ell \\ &\le r^{-(n-1)}(1/3)^{4n} = r\left(\frac{\sqrt{15}}{27}\right)^n.\end{aligned}$$

The first-1 prefix $0^{j-1}1$ has power weight $u^{j-1}v$. Applying the cylinder bound and summing over $j \le 4n$ gives the claimed second term because

$$\sum_{j=1}^{4n} u^{j-1}v \le \frac{v}{1-u} = 4 + \sqrt{15}.$$

This argument holds for every source in $\mathcal{Q}$, including the source ν_n used above. To obtain the simpler constant in (7), put $\alpha = \sqrt{15}/27$, $\beta = 225/256$, and $c = (4 + \sqrt{15})/(3\sqrt{15})$. The bound is $\beta^n + c\alpha^n$. Since $\alpha < \beta$ and $c\alpha/\beta = 256(4 + \sqrt{15})/18225 < 1/8$, it is at most $(9/8)\beta^n < 1$ for every $n \ge 1$.

For the lower bound in (7), take an i.i.d. source with probabilities u and $1 - u$. Both are at most the corresponding square-root reference probabilities, since $1 - u \le v$. It belongs to $\mathcal{Q}$, and its prefix 0^{8n} leaves B unfinished under every selector. Its probability $u^{8n} = (15/16)^{4n}$ is therefore a lower bound for every selector, including the minimax optimum.

The selector invariance of the stopping law and all prefix power sums follows from Theorem 3.1, applied to these success sets. □

F Proof for noisy predecessor search

Proof of Theorem 5.1. Set $x_0 = -\infty$ and $x_{n+1} = +\infty$. Leaf i represents $[x_i, x_{i+1})$ and returns index i, with zero meaning no predecessor. Internal nodes represent intervals $[x_i, x_j)$ and split at x_k, $i < k < j$. Chain nodes inherit their leaf's interval and label. These half-open intervals handle equality using strict queries.

At a nonterminal state with interval $[x_i, x_j)$, use the lower and upper comparisons $x < x_i$ and $x < x_j$ when their boundaries are finite. Verification requires the lower answer to be false and the upper answer to be true. An infinite boundary is known exactly; replace its comparison by a dummy comparison $x < x_1$ and ignore the answer. At an original internal node the third comparison is its split comparison $x < x_k$; at an original leaf or chain node it is another dummy $x < x_1$. Each of these three logical comparisons is answered by the majority of exactly ℓ repetitions of the same primitive query. All 3ℓ queries are made before the move, including dummy queries and queries after an earlier majority already makes verification fail. On failure move to the parent, with a requested upward move at the root replaced by staying there. On success take the split-selected child at an internal node and the unique downward chain edge otherwise. At a chain endpoint return immediately, without further queries.

Let z be the correct original leaf and z^* its chain endpoint. If a node's interval excludes x, three correct majorities make verification fail, and its parent is the neighbor toward z^*. If its interval contains x, correct answers select the child toward z^*, or advance on z's chain. Thus every raw block in G moves one edge toward z^*. Every other block moves at most one edge. For live states use the positive integer potential $d(v) = \mathrm{dist}(v, z^*)$, initially $d_0 = 2h + L_\eta$; assign potential zero on termination.

For odd ℓ, define the weight of a correct majority by

$$M_s = \sum_{j>\ell/2} \binom{\ell}{j} p^{sj}(1-p)^{s(\ell-j)}.$$

Since $p > 1/2$, the law of large numbers ensures an odd ℓ with $M_1 > 2^{-1/3}$. Choose the least such ℓ, so it depends only on p. Group raw errors into consecutive blocks of length 3ℓ, with reference mass $\pi(e) = p^{3\ell-|e|_1}(1-p)^{|e|_1}$. Equation (14) directly bounds every length-T block prefix by D times its reference mass to power s; no change in the primitive source assumption is made. Summing over raw patterns gives

$$a_s = M_s^3, \qquad b_s = (p^s + (1-p)^s)^{3\ell} - M_s^3.$$

Both are sums of $\pi(e)^s$ over fixed nonempty sets, so both strictly decrease in s. At $s = 0$, their product is $7 \cdot 2^{6\ell-6} > 1/4$; at $s = 1$, it is $M_1^3(1 - M_1^3) < 1/4$. This proves existence and uniqueness of s_c. Also $a_{s_c} \geq M_1^3 > 1/2$, so $a_{s_c} = \lambda/2$, $b_{s_c} = 1/(2\lambda)$, and $\lambda > 1$. This standard amplification is analyzed on raw patterns: the correct weight is M_s^3, not M_1^{3s}. No source condition on the majority answers is assumed.

For every nonnegative increasing f with $f(0) = 0$, the live children of a history of potential d satisfy

$$\sum_{e:\,\text{live}} \pi(e)^{s_c} f(d_e) \leq a_{s_c} f(d-1) + b_{s_c} f(d+1). \tag{38}$$

This is the complete one-step input to the proof of Lemma C.1: its tree induction uses powered edge masses, so the same proof applies to these nonuniform edges. In particular, for any integer $m > d_0$, the weight of prefixes first reaching m before termination is at most

$$H_m(d_0) = \frac{d_0}{m}\lambda^{d_0-m}, \qquad H_m(d) = \frac{d}{m}\lambda^{d-m}.$$

Indeed, $H_m(0) = 0$, $H_m(m) = 1$, and $a_{s_c}H_m(d-1) + b_{s_c}H_m(d+1) = H_m(d)$; expanding the finite tree to any depth bounds its first-hit prefixes by $H_m(d_0)$, and increasing the depth gives the infinite-horizon bound. The remaining live prefixes confined below m have powered mass at most

$$\frac{\lambda^{d_0}}{\sqrt{\lambda^2-1}} \exp\left(-\frac{\pi^2 T}{2(2m-1)^2}\right),$$

by the same reflected-strip comparison and diagonal conjugation in that lemma. Multiplying the two cover sums by D and optimizing m therefore gives

$$\Pr_\nu\{Q > 3\ell T\} \leq \min\{1, C_p D \lambda^{d_0} e^{-c_p T^{1/3}}\}, \qquad \mathbb{E}_\nu Q \leq C_p(d_0 + \log D + 1)^3.$$

The inclusion of source classes gives this bound for all $s \geq s_c$. For $s > s_c$, since $\max_e \pi(e) = p^{3\ell}$,

$$\frac{a_s}{\lambda} + b_s\lambda \leq p^{3\ell(s-s_c)}\left(\frac{a_{s_c}}{\lambda} + b_{s_c}\lambda\right) = p^{3\ell(s-s_c)}.$$

Finite-tree induction on the powered sum of λ^d over live depth-T prefixes yields $\Pr_\nu\{Q > 3\ell T\} \le \min\{1, D\lambda^{d_0} p^{3\ell(s-s_c)T}\}$. Summing and counting 3ℓ queries per move gives $C_p(d_0 + \log D + 1)/(s - s_c)$. Since $d_0 = 2h + L_\eta = O_p(K)$, these bounds prove the query claim and almost-sure termination.

Finally consider an incorrect leaf u of depth d_u, and let c be the depth of its lowest common ancestor with z. Its chain endpoint u^* satisfies

$$\operatorname{dist}(u^*, z^*) = 2h + 2d_u - 2c + 2L_\eta \ge d_0 + L_\eta + 2.$$

Set $m = d_0 + L_\eta + 1$. Any run terminating at u^* must first reach distance m at a live state: its final endpoint has distance at least $m + 1$, and each move changes distance by at most one. Consequently the first-hit bound, applied before resetting the terminal potential to zero, gives

$$\Pr_\nu\{\text{incorrect predecessor}\} \le DH_m(d_0) \le D\lambda^{-L_\eta - 1} \le \eta.$$

Only the algorithm's visited leaf label is returned; the unknown correct endpoint is used solely in this analysis. □

G Composition under a shared source

Sharing a random stream can alter robustness even when a selector may interleave components arbitrarily. The following process gives an exact comparison between a prefix probability bound on the shared tape and separate bounds on each component's subsequence.

Each component is a connected N-vertex graph with binary vertex values. Select an edge whose endpoints disagree. Four source labels replace its endpoint values by 00, leave them unchanged, reverse them, or replace them by 11. The reference probabilities are $(49/100, 1/4, 1/4, 1/100)$, respectively. Constant components are finished and never selected again. This extends the discordant voting model of Cooper, Dyer, Frieze, and Rivera [CDFR18].

Let $\Delta_N \in (1/2, 1)$ be the unique solution of

$$2\,4^{-\Delta_N} + 2(7/100)^{\Delta_N} \cos(\pi/N) = 1.$$

Corollary G.1 (Small individual failure, certain joint failure)**.** *For every $N \ge 3$, start each of $N - 1$ components with exactly one vertex equal to one and all others zero. In isolation, each has worst-case nontermination probability*

$$\max_{\nu \in \mathcal{Q}_{\Delta_N, 1}} \nu\{\tau_i = \infty\} = 7^{-\Delta_N (N-2)} = \exp[-\Theta(N)].$$

There is one source in $\mathcal{Q}_{\Delta_N,1}$ under which their combined execution never terminates, simultaneously for every deterministic nonanticipating rule that chooses a disagreeing edge. The rule may interleave components arbitrarily and remember all past labels. Under the independent reference source, every such rule has expected total running time at most $25(N - 1)/12$.

The following formula proves the corollary and also allows components of different sizes and arbitrary initial assignments.

Theorem G.2 (Exact robustness under arbitrary interleaving)**.** *Fix $N \geq 3$, put $s = \Delta_N$, $\theta = \pi/N$, and $r = 7^s$, and define*

$$V_N(k) = \begin{cases} 0, & k = 0, \\ r^{k-N+1}\dfrac{\sin(k\theta)}{\sin\theta}, & 1 \leq k < N-1, \\ 1, & k \geq N-1. \end{cases}$$

Let the graph be a disjoint union of finitely many connected components, each with at least N vertices. Let K_0 count the ones in the initially nonconstant components. For every deterministic nonanticipating edge selector and every $D \geq 1$,

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu\{\tau = \infty\} = \min\{1, DV_N(K_0)\}. \tag{39}$$

One source depending only on N, K_0, D attains this value simultaneously for every allowed graph, initial assignment with count K_0, and selector. Under the independent reference source,

$$\mathbb{E}_P \tau \leq \frac{25}{12} K_0. \tag{40}$$

Proof. Write $V = V_N$ and let

$$d = (49/100)^s, \qquad b = 2(1/4)^s, \qquad u = (1/100)^s, \qquad a = \sqrt{du} = (7/100)^s.$$

The equation defining s is $b + 2a\cos\theta = 1$. Its left side is strictly decreasing in s, exceeds one at $s = 1/2$, and is below one at $s = 1$. Thus $1/2 < s < 1$. In particular, $b < 1$, $d = ar$, $u = a/r$, and $r > \sqrt{7} > 2\cos\theta$.

Set $h(k) = r^k \sin(k\theta)$ for $0 \leq k \leq N$. The sine addition formula gives

$$dh(k-1) + bh(k) + uh(k+1) = h(k), \qquad 1 \leq k < N.$$

Moreover, h is strictly increasing on $1, \ldots, N-1$: for $1 \leq k \leq N-2$, $\sin(k\theta) \leq 2\cos\theta \sin((k+1)\theta)$ and $r > 2\cos\theta$. Hence $0 < V(k) < 1$ for $1 \leq k < N-1$, and the same harmonic identity holds for V there. At the upper boundary it gives

$$dV(N-2) + b = 1. \tag{41}$$

At any history, let K count the ones in components that are still nonconstant. A downward action sends K to $K-1$; either neutral action leaves it unchanged. An upward action sends K to $K+1$, unless it makes an L-vertex component all ones, in which case K becomes $K+1-L$. Such a completion requires $K \geq L-1 \geq N-1$. Thus below $N-1$ the count has the same increments for every selector, even though this need not hold above that level.

For the upper bound, only $1 \leq K_0 < N-1$ requires proof. Cover each nonterminating tape at its first visit to $K = N-1$, or at depth T if it has not yet visited that level. Before the first visit, the powered count matrix on $1, \ldots, N-2$ is tridiagonal with lower diagonal d, diagonal b, and upper diagonal u. Call it A. Diagonal similarity to a symmetric tridiagonal matrix gives

$$\rho(A) = b + 2a\cos\frac{\pi}{N-1} < 1.$$

The harmonic identity with boundary values $V(0) = 0$ and $V(N-1) = 1$ therefore makes the total weight of first visits to $N-1$ equal to $V(K_0)$. The remaining live prefixes at depth T have weight

$e_{K_0}^{\mathsf{T}} A^T \mathbf{1}$, which tends to zero. Applying the prefix bound to this finite cover and letting $T \to \infty$ gives $\nu\{\tau = \infty\} \le \min\{1, DV(K_0)\}$.

For the matching source, suppose $K_0 > 0$. An auxiliary law q maintains a counter $Y_0 = K_0$. Let $\delta(z) \in \{-1, 0, 1\}$ be the increment assigned to action label z. When $1 \le Y_t < N - 1$, use

$$q(z \mid Y_t) = \pi(z)^s \frac{V(Y_t + \delta(z))}{V(Y_t)}.$$

When $Y_t \ge N - 1$, give each neutral label probability $(1/4)^s$, the downward label probability $1 - b$, and the upward label probability zero. Update $Y_{t+1} = Y_t + \delta(z)$. The harmonic identity normalizes these probabilities below $N - 1$, and normalization above that level is immediate. The downward probability at one is zero, so Y_t stays positive.

Every upward move starts below $N-1$, so it cannot finish a component. A downward completion produces an all-zero component and has no additional effect on the count. Consequently $K_t = Y_t$ for every graph, initial assignment, and selector under consideration. Every tape in the support of q is nonterminating for all these processes.

Equation (41), together with $V(N - 2) < 1$, gives $1 - b = dV(N - 2) < d$. It follows in the upper range as well as the lower range that

$$q(z \mid Y_t) \le \pi(z)^s \frac{V(Y_{t+1})}{V(Y_t)}.$$

Multiplying along a prefix yields

$$q[w] \le P[w]^s \frac{V(Y_{|w|})}{V(K_0)} \le \frac{P[w]^s}{V(K_0)}. \tag{42}$$

Put $p = \min\{1, DV(K_0)\}$. The subprobability law $\eta = pq$ has mass p and obeys $\eta[w] \le DP[w]^s$. It extends to a probability law ν obeying the same bound: at each node, distribute its assigned mass among the children, preserving their η masses and respecting their capacities $DP[wz]^s$. This is possible because the lower bounds sum to $\eta[w]$, whereas the capacities sum to $DP[w]^s \sum_z \pi(z)^s \ge DP[w]^s$. Starting with root mass one and using a fixed label order gives an extension depending only on N, K_0, D. Its common nontermination mass is at least p, proving equality in (39). If $K_0 = 0$, all components are already constant and the assertion is immediate. When $K_0 \ge N - 1$ and $D = 1$, $p = 1$ and q itself is the required source.

Finally, under P, a live update has downward drift at least $49/100 - 1/100 = 12/25$ in K; finishing an all-one component only increases that downward drift. Summing the conditional drift bound up to $\tau \wedge t$ gives $(12/25)\mathbb{E}_P(\tau \wedge t) \le K_0$. Letting $t \to \infty$ proves (40). □

For one component starting with one vertex equal to one, $V_N(1) = 7^{-\Delta_N(N-2)}$. For $N - 1$ such components, $V_N(N - 1) = 1$ and the same source forces nontermination under every interleaving. Equation (40) gives the reference mean bound. This proves Corollary G.1.

AI methodology statement

The author posed the research questions, chose the source model and the three settings, and directed the investigation step by step, selecting at each stage which of several candidate directions to pursue; this led successively to the scheduling separation, the sharp threshold and cubic rate for the SAT walk, the restart criterion, and the noisy-search guarantee, and the author supplied the interpretation of these results as one source model with three applications. Throughout this process the author used ChatGPT and Codex (OpenAI) as research aids; their assistance included exploring and drafting proof arguments, comparing related literature, and revising exposition, and it materially influenced Sections 2–6 and the associated appendices. Claude (Anthropic) assisted with the organization and exposition of the abstract and main text, including the wording and structure of several theorem statements. The author evaluated the proposed arguments, checked the proofs and cited sources, revised the material retained in the paper, and made the final decisions about its claims and presentation. The author verified the correctness and originality of all content, including references, and takes full responsibility for the paper's accuracy, attribution, and integrity.

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